Project Report On

Blockchain Based Incident Reporting System

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

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CERTIFICATE

This is to certify that the project report entitled ***"Blockchain Based Incident Reporting System"*** *is a bonafide record of the work done by* ***Siona Kurisinkal Biju (U2203194), Treesa Maria Cinil (U2203205), Susen Ann George (U2203202), Niya Sony (U2203167)****, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

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Abstract

Rapid urbanization has increased the need for dependable and open systems for managing and reporting incidents like accidents, water and sanitation issues, infrastructural damages, etc. In order to facilitate real-time reporting of such incidents, this paper introduces a Decentralized Incident Reporting System (IRS) based on Blockchain technology. The proposed system allows citizens to share geolocation data, details of the event, and image based evidence through a web-based interface. By ensuring immutability and transparency of such records, Blockchain technology promotes accountability and trust. To reduce storage overhead, images of the reported issues are stored on the InterPlanetary File System (IPFS), and smart contracts maintain important metadata and verification status on-chain. MetaMask wallets are used to implement role-based access control. The system features an interactive visualization mechanism based on Leaflet and OpenStreetMap, as well as a verification procedure controlled by smart contracts. An AI-assisted chatbot is also available that answers basic user queries and helps identify incidents from uploaded images. The proposed IRS's modular and decentralized architecture promotes citizen participation, improves incident traceability, and facilitates scalable smart city governance.

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List of Abbreviations

IRS	Incident Reporting System
AI	Artificial Intelligence
IPFS	InterPlanetary File System
API	Application Programming Interface
GUI	Graphical User Interface
OSM	OpenStreetMap
DB	Database
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
CID	Content Identifier
CIRS	Critical Incident Reporting Systems
EMS	Emergency Medical Services
GPS	Global Positioning System
WGT	Water Game Token
ICO	Initial Coin Offering
URL	Uniform Resource Locator
ERC	Ethereum Request for Comments

Chapter 1

Introduction

Traditional incident reporting mechanisms often face issues such as lack of transparency, delayed response, and centralized control. With the rapid growth of urban environments, there is an increasing need for reliable and efficient systems that allow citizens to report incidents such as accidents, infrastructure damage, and public service issues in a timely manner.A decentralized approach using Blockchain technology helps ensure data integrity, transparency, and accountability by maintaining tamper-proof records of reported incidents. The system aims to enable secure and real-time reporting while improving communication between citizens and authorities. It also defines the problem context, outlines the objectives of the system, and highlights the challenges and assumptions considered during the design and implementation. Additionally, the relevance of the system in supporting transparent and efficient urban governance is emphasized, followed by an overview of how the report is structured.

1.1 Background

In rapidly growing urban areas, incidents such as traffic accidents, road blockages, flooding,and safety hazards occur frequently, requiring quick and reliable reporting mechanisms. Traditional incident reporting systems are centralized, which makes them prone to data manipulation, delayed responses, and limited transparency, ultimately reducing public trust in authorities. Citizens often lack a unified platform to report and track incidents in real time. With the emergence of Blockchain technology, it has become possible to create systems that are decentralized, transparent, and tamper-proof. Blockchain ensures that once data is recorded, it cannot be altered or deleted. This maintains data integrity and public accountability. This project leverages the potential of Blockchain to build a secure, decentralized incident reporting system, enabling citizens to report incidents transparently. It also empowers authorities to verify and respond efficiently, contributing to the development of smart and trustworthy urban governance.

1.2 Problem Definition

Existing incident reporting systems are centralized, making them vulnerable to data manipulation, delays, and lack of transparency. This reduces public trust and slows emergency responses. There is no secure, real-time, and verifiable platform that allows citizens to report and track urban incidents transparently.

1.3 Scope and Motivation

1.3.1 Scope

The scope of this project is to design and develop a Blockchain-Based Incident Reporting System that enables citizens to securely report, verify, and track incidents occurring in urban areas. The system utilizes Ethereum Blockchain and smart contracts to ensure that all reported data remains immutable and transparent. It provides a web-based interface integrated with MetaMask for user authentication and Leaflet maps for real-time visualization of incidents. The project also includes a verification module for authorized personnel to review and validate reports through Blockchain transactions. By implementing a decentralized architecture, the system aims to enhance reliability, accountability, and efficiency in urban incident management.

1.3.2 Motivation

The motivation behind this project arises from the limitations of traditional incident reporting systems, which are often centralized and lack real-time verification. In many cities, data transmission delays and lack of transparency hinder quick responses and reduce citizen confidence in reporting mechanisms. The adoption of Blockchain technology presents an opportunity to create a more secure system that guarantees data integrity and public transparency. By allowing citizens to directly report and verify incidents through a decentralized platform, the project fosters greater civic participation and improves collaboration between the public and authorities. Ultimately, the system is motivated by the goal of contributing to safer and more accountable urban governance.

1.4 Objectives

- To develop a decentralized web platform using Ethereum Blockchain for secure and transparent urban incident reporting.
- To implement smart contracts that automate the processes of incident submission, verification, and storage on the Blockchain.
- To provide a user-friendly web interface integrated with MetaMask for authentication and incident submission.
- To enable real-time visualization of incidents on an interactive map using Leaflet API with status-based color indicators.
- To improve urban management through real-time incident data for better response and planning.

1.5 Challenges

The project faces challenges such as Blockchain transaction delays, gas fees, and network congestion, which can impact the speed of incident reporting and verification. It must also ensure smooth user adoption by simplifying MetaMask integration and preventing false reports through strong verification mechanisms. Additionally, scalability, increasing storage costs, and maintaining verifier accountability are critical for long-term reliability and trust.

1.6 Assumptions

The system assumes that users have a stable internet connection and a compatible web browser that supports MetaMask integration. It also relies on the Ethereum Blockchain network functioning properly with reasonable transaction fees and minimal congestion during reporting and verification. It is expected that users and verifiers provide accurate information and act responsibly while submitting and validating incident reports.

1.7 Societal and Industrial Relevance

This project holds significant relevance for both society and industry by enhancing transparency, trust, and accountability in urban incident management through Blockchain technology

- **Societal Impact:** Empowers citizens to actively participate in urban governance through transparent and tamper-proof reporting of incidents such as accidents, road damage, and safety hazards.
- **Transparency and Trust:** Eliminates centralized control and ensures data integrity and public confidence through decentralized Blockchain verification.
- **Industrial Applicability:** Assists government bodies, municipal corporations, and smart city authorities in efficient monitoring, analysis, and decision-making using reliable real-time data.
- **Technological Advancement:** Showcases the integration of Blockchain, geolocation, and smart contracts for secure, automated, and auditable data management.
- **Scalability and Replicability:** Features a modular architecture that can be extended to other domains like healthcare reporting, supply chain tracking, and environmental monitoring.

1.8 Organization of the Report

This report is structured as follows:

- **Chapter 1: Introduction** presents the background of the study, problem definition, objectives, scope, challenges, assumptions, and the significance of the proposed system.
- **Chapter 2: Literature Survey** reviews existing research and prior work related to blockchain-based incident reporting systems and related technologies.
- **Chapter 3: System Design** describes the architecture of the proposed system, its main components, methodologies used, and overall system workflow.

- **Chapter 4: Results and Discussion** presents the experimental evaluation of the system, performance metrics and testing results.
- **Chapter 5: Conclusion** summarizes the key findings of the project and outlines possible future improvements.

This structured organization ensures a clear and logical flow of information, guiding the reader from the conceptual foundation to the technical implementation and outcomes of the project. It provides a comprehensive understanding of how Blockchain technology is applied to enhance transparency and efficiency in urban incident reporting.

1.9 Summary of Chapter

The first chapter introduces the Blockchain-Based Incident Reporting System, emphasizing its purpose, scope, and significance in addressing inefficiencies in traditional, centralized reporting mechanisms. It highlights how Blockchain technology ensures data integrity, transparency, and citizen empowerment in urban incident management. The chapter defines the core problem of unreliable and opaque reporting systems and establishes the objective of creating a decentralized web platform using Ethereum and smart contracts for secure, tamper-proof reporting. It outlines the system's scope, motivation, and technical components such as MetaMask authentication and Leaflet-based visualization. Key challenges related to Blockchain scalability, transaction delays, and data verification are discussed along with necessary assumptions for system operation. Finally, the societal and industrial relevance of the project is emphasized, showing its potential to enhance public trust and governance efficiency after which the report structure is outlined to guide readers through the subsequent chapters.

Chapter 2

Literature Review

2.1 Introduction

The increasing demand for transparency, security, and accountability in urban governance has accelerated research into decentralized reporting frameworks powered by Blockchain technology. Recent developments combine smart contract automation, decentralized storage solutions like IPFS, and geolocation-based data integration to create tamper-proof, community-driven reporting systems. By leveraging Blockchain's immutability and transparency, such frameworks eliminate centralized control and ensure data integrity throughout the incident lifecycle. Smart contracts facilitate trusted interactions among citizens, verifiers, and authorities, while off-chain storage optimizes scalability and performance. The following sections review core design principles, algorithms, and technological choices that underpin the proposed blockchain-based incident reporting framework, highlighting its potential to enhance urban safety, citizen engagement, and data-driven decision-making.

2.2 Existing Techniques

2.2.1 Decentralized Incident Reporting: Mobilizing Urban Communities with Blockchain[1]

Problem Addressed

Traditional incident reporting systems in urban environments are centralized, resulting in delayed responses, data manipulation, and reduced transparency. They often rely on intermediaries or authorities, which diminishes public trust and citizen engagement, and lack a unified, tamper-proof platform for real-time reporting, verification, and tracking of incidents. Existing digital solutions fail to guarantee data integrity, as reports can be altered or deleted without traceability, and managing large-scale urban data is challenging due to scalability and privacy constraints in centralized architectures. These limitations underscore the need for a decentralized, transparent, and verifiable framework that leverages Blockchain and IPFS to provide secure, efficient, and reliable incident reporting.

Objective

- Develop a decentralized, Blockchain-based framework for urban incident reporting that ensures transparency, immutability, and public trust.
- Utilize smart contracts on a permissioned Blockchain (Quorum) to securely record, verify, and manage incident data without relying on centralized authorities.
- Integrate IPFS (InterPlanetary File System) for efficient off-chain storage of multimedia evidence while maintaining verifiable links on-chain.
- Enable real-time geolocation-based reporting and tracking of incidents to enhance urban safety and responsiveness.
- Empower citizens to actively participate in community-driven incident reporting, improving accountability and trust between the public and authorities.
- Achieve scalability, low latency, and enhanced data privacy through architectural optimization and permissioned consensus mechanisms.

Methodology

The proposed system employs a modular, three-phase architecture combining Blockchain, decentralized storage, and geolocation-based visualization to ensure secure, transparent, and real-time incident reporting.

1. Incident Reporting Phase

- Users can report urban incidents such as accidents, hazards, or traffic congestion through a web interface.
- Each report includes essential metadata location, description, timestamp, and one or more media files (images or videos).

- Uploaded media files are stored in IPFS (InterPlanetary File System), generating a unique Content Identifier (CID) for each file.
- A smart contract deployed on the Ethereum-based Quorum Blockchain records the incident metadata along with the IPFS CID, ensuring immutability and traceability.

2. Verification and Status Update Phase

- Authorized verifiers, such as municipal officers or trusted community representatives, can review pending reports.
- Verifiers access detailed report data, including the IPFS media link, and can update the status to *Verified* or *Resolved*.
- Each status update triggers a Blockchain transaction, preserving a permanent audit trail of every modification while maintaining full transparency.

3. Visualization and Retrieval Phase

- All reported incidents are dynamically retrieved from the Blockchain and visualized on an interactive map using APIs such as Leaflet.js or Mapbox.
- Markers are color-coded to represent report status:
  - Green – Verified
  - Yellow – Pending
  - Blue – Resolved
- Citizens and authorities can view incidents in real time, fostering rapid response and improved situational awareness.

This methodology ensures a transparent and tamper-proof system where every incident is traceable, verifiable, and securely stored. The integration of IPFS for off-chain storage enhances scalability, while the permissioned Quorum Blockchain minimizes latency and ensures efficient consensus.

Algorithms

The proposed system uses three core algorithms to handle the reporting, visualization, and verification of incidents. Each algorithm operates in coordination with the Blockchain network and IPFS to ensure transparency, immutability, and security.

Algorithm 1: Incident Reporting : Submit a new incident report with location, description, and media to the Blockchain and IPFS.

1. Start when the user selects *"Report Incident"* from the main menu.
2. Prompt the user to:
   - Enter location.
   - Enter incident description.
   - Upload at least one media file (photo).
3. Validate inputs:
   - Location is successfully captured.
   - Description is not empty.
   - At least one media file is uploaded.
4. Upload the media file(s) to IPFS using an IPFS API or node.
5. Retrieve the CID (Content Identifier) from IPFS after upload.
6. Prepare metadata:
   - `incident_id` (auto-generated)
   - `timestamp` (server-side or Blockchain)
   - `location coordinates`
   - `IPFS CID`
   - `status = Pending`
7. Call the Blockchain smart contract function: `reportIncident(incident_id, location, timestamp, CID, status)`.

8. On success, display a confirmation message: "Incident submitted successfully and is now pending verification."

9. End.

Algorithm 2: Incident Viewing : Fetch all incidents from the Blockchain and display them on a map with their current status.

1. Start when the user selects *"View Incidents"* from the main menu.

2. Call the Blockchain smart contract function: `getAllIncidents()`, which returns a list of incidents with metadata (location, description, CID, status).

3. For each incident:

   - Retrieve media preview link from IPFS using the CID.
   - Assign a map marker color:
     - Green – Verified
     - Yellow – Pending

4. Display markers on the map using `Leaflet.js` or `Mapbox`.

5. When the user clicks a marker, show a popup with:

   - Description
   - Status
   - Timestamp
   - Link to IPFS media

6. End.

Algorithm 3: Incident Verification : Allow authorized verifiers to update the status of pending incidents.

1. Start when the verifier logs in with MetaMask and accesses the Verifier Dashboard.

2. Fetch all incidents from the Blockchain with `status = Pending`.

3. For each incident:
   - Display details: location (map), description, IPFS media link.
4. Verifier selects an incident and chooses one of the following:
   - `Verify` → status = Verified
   - `Resolved` → status = Resolved
5. Call the Blockchain smart contract function: `updateIncidentStatus(incident_id, new_status)`.
6. The Blockchain appends the status update transaction, keeping the original report intact.
7. The updated status is visible immediately on the public map.
8. End.

2.2.2 Critical Incident Reporting System (CIRS): A Fundamental Component of Risk Management in Health Care Systems to Enhance Patient Safety.[2]

Problem Addressed

The complexity of health care systems, increasing clinical demands, and the introduction of novel medical technologies have rendered hospitals high-risk environments, where patients are susceptible to errors and adverse events. Despite ethical and legal obligations to ensure patient safety, the lack of a well-established error-reporting culture and standardized reporting systems in many countries, including Germany and Austria, limits the identification and mitigation of risks. Critical Incident Reporting Systems (CIRS) are often implemented intuitively rather than systematically, leading to inconsistent reporting, underutilization by medical professionals, and limited insights into preventable errors. Consequently, there is a pressing need to assess the impact of CIRS on clinical risk management and patient safety and to identify best practices for effective implementation.

Objective

The objectives of this study are:

- To provide a comprehensive overview of Critical Incident Reporting Systems (CIRS) and their utility in clinical risk management.
- To examine institutional processes for implementing CIRS, offering guidance on achieving efficient and sustainable adoption in healthcare organizations.
- To investigate the relationship between CIRS adoption and patient safety, highlighting factors that enhance or impede the effectiveness of incident-reporting systems.

Methodology

The CIRS framework consists of three main components: Incident Reporting, Incident Management, and Data Analysis.

1. Incident Reporting Healthcare personnel can report incidents via a structured interface, providing relevant details such as type, severity, location, and contributing factors. The system supports both manual and semi-automated reporting, ensuring completeness and standardization of data.

2. Incident Management Reported incidents are stored in a centralized database with secure access controls. The system tracks the status of each incident, assigns responsibilities for investigation, and maintains an audit trail for accountability. Alerts and notifications ensure that critical incidents receive prompt attention.

3. Data Analysis and Risk Management CIRS applies statistical and trend analysis to the reported incidents to identify recurring patterns and high-risk areas. This enables healthcare administrators to implement preventive measures, design training programs, and update protocols. Insights from the system contribute to continuous quality improvement and enhanced patient safety.

2.2.3 Deployment of Critical Incident Reporting System (CIRS) in Public Styrian hospitals: A Five year Perspective [3]

Problems Addressed

Organizations often face several challenges in managing incident reporting effectively. There is a tendency for underreporting of incidents due to fear of blame or lack of awareness among individuals. Additionally, aggregating and analyzing data from multiple departments becomes difficult, leading to fragmented insights. These inefficiencies result in delays when identifying recurring patterns or systemic risks. Ensuring accurate and consistent information across various reporting sources further adds to the complexity. Moreover, prioritizing and following up on critical incidents often becomes inefficient, hindering timely resolution and overall organizational safety.

Objectives

- Implement a system that allows healthcare staff to easily report incidents in a timely and structured manner.
- Encourage a non-punitive reporting culture to improve reporting rates.
- Aggregate and analyze incidents to identify patterns, high-risk areas, and preventive measures.
- Ensure consistent, accurate, and complete documentation of incidents across departments.
- Provide actionable insights and alerts to healthcare management for prompt intervention and continuous improvement of patient safety.

Methodology

The Critical Incident Reporting System (CIRS) is designed to collect, document, and analyze patient safety incidents across healthcare facilities. It facilitates timely reporting, maintains a database of incidents, and supports proactive interventions.

System Components

- **Incident Reporting Interface**: Allows healthcare professionals to log incidents easily, with structured fields capturing relevant details.
- **Data Management and Storage**: Centralized database to store incident reports securely, ensuring confidentiality and accessibility.
- **Analysis and Risk Identification**: Tools to analyze trends, detect patterns, and identify high-risk processes or departments.
- **Alerts and Feedback Mechanism**: Provides real-time notifications and feedback to relevant stakeholders for immediate corrective actions.
- **Reporting and Documentation**: Generates reports for management review, regulatory compliance, and continuous quality improvement.

2.2.4 A Survey on Incident Reporting and Management systems[4]

Problems Addressed

Existing incident reporting systems often suffer from several limitations that affect their effectiveness and reliability. Many lack robust mechanisms to ensure complete and accurate reporting, resulting in data gaps and inconsistencies. The absence of anonymous reporting options can further discourage users from submitting sensitive incidents, reducing overall participation. Additionally, challenges related to data security and the prevention of false reports continue to persist across both public and private reporting platforms. Furthermore, the limited availability of comparative analyses on system features and performance makes it difficult to identify best practices and develop standardized, efficient reporting frameworks.

Objectives

- Analyze and compare incident reporting systems based on technical features, usability, performance, security, and anonymity.
- Identify strengths and weaknesses of existing systems to guide future design and implementation of robust reporting solutions.

- Highlight challenges in preventing false reports, ensuring data security, and supporting anonymous reporting.

Methodology

- **System Survey & Feature Identification**: Collected and reviewed existing mobile and web-based incident reporting applications, analyzing features such as platform support, availability, usability, performance, GPS/location tracking, reliability, security, and anonymity.
- **Comparative Evaluation**: Classified systems into Public (non-anonymous) and Private (anonymous) categories, comparing them via user ratings, app store reviews, documentation, and experimental results (e.g., Dijkstra's algorithm in SPEARS, ring signatures in privacy-preserving EMS).
- **Analysis & Ranking**: Assessed technical strengths and weaknesses of each system, identifying best-performing solutions (e.g., Ushahidi, Privacy-preserving EMS) and highlighting common issues such as lack of reliability, weak data security, or limited anonymous reporting options.

Key Findings

- Ushahidi outperforms other public systems in performance, reliability, usability, and partial data security.
- Most systems are free, GPS-enabled, and user-friendly.
- Many systems still lack robust reliability mechanisms, data security safeguards, or anonymous reporting capabilities, indicating areas for future improvement.

2.2.5 A Blockchain Approach for Migrating a Cyber-Physical Water Monitoring Solution to a Decentralized Architecture.[5]

Problem Addressed

Traditional water monitoring systems face several inherent challenges due to their centralized design and limited incentive structures. These systems typically rely on centralized databases and client-server architectures, where only service providers have direct control over the information, resulting in reduced transparency and minimal user participation. Additionally, existing crowdsensing platforms use centralized reward systems that restrict users from monetizing or trading their earned rewards outside the application ecosystem. The lack of transparency in reward distribution and discount application processes often necessitates manual administrative intervention, leading to inefficiencies. Moreover, the incentive mechanisms in such systems remain static and non-tradeable, failing to adapt dynamically or provide users with meaningful, transferable benefits both within and beyond the platform.

Objectives

- **Migrate to Decentralized Architecture**: Transform the existing centralized cyber-physical water monitoring solution to a hybrid architecture integrating Blockchain technology.

- **Tokenize Rewards**: Implement ERC-20 tokens as rewards for incident reporting in urban water infrastructure, enabling users to trade tokens both within and outside the application.

- **Digitalize Discounts**: Create ERC-1155 tokens representing subscription discounts that can be purchased, sold, and automatically applied without manual intervention.

- **Enhance Transparency and Trust**: Leverage Blockchain to increase solution security, transparency, and trust between water providers and clients through immutable transaction records.

- **Improve User Engagement**: Increase customer involvement in water reporting flows by offering tradeable rewards and dynamic subscription discounts.

Methodology and Architecture

The proposed solution migrates the baseline crowdsensing water monitoring system to a decentralized architecture using Ethereum Blockchain technology.

- **ERC-20 Token Implementation (WGT)**: Custom Water Game Token deployed on Ethereum Blockchain; serves as primary reward mechanism; transferable and tradeable on external platforms.
- **Smart Contracts**:
  - *Rewarder Contract*: Manages token distribution as rewards; mintable supply with admin controls; pause/start functionality for critical issues.
  - *Donor Contract*: Facilitates token donations between users; wrapper over ERC-20 with sender-paid transaction fees.
  - *TokenSale Contract*: Implements Initial Coin Offering (ICO) mechanism; mintable supply with admin controls.
  - *Discount Contract*: ERC-1155 multi-token contract for subscription discounts; supports both fungible and non-fungible assets.
  - *Marketplace Contract*: Wrapper facilitating discount purchases; enables trading WGT tokens for discount tokens.
- **Decentralized Storage**: IPFS (via Pinata) stores ERC-1155 metadata and discount images; maintains discount properties (name, description, value, image URL).
- **Hybrid Architecture Flow**:
  1. User reports incident via crowdsensing application
  2. Incident stored in relational database
  3. WGT tokens transferred from admin wallet to user wallet via Blockchain
  4. User purchases discount from marketplace using WGT tokens

5. ERC-1155 discount token issued to user
6. Physical discount automatically updated in database
7. System continuously scans for digital discounts and synchronizes with database

- **Integration Layer**: MetaMask wallet integration for Blockchain authentication; retrieves user account information and balance; facilitates transaction signing and execution.
- **Automatic Discount Application**: Smart contracts monitor user-owned ERC-1155 tokens; automatically apply/remove discounts based on token ownership; eliminates manual administrative intervention.
- **Secondary Market Trading**: Users can trade both WGT tokens and discount tokens on external cryptocurrency exchanges; subscription discounts automatically adjusted based on token ownership changes.

2.3 Comparison of Related Works

2.3.1 Gaps Identified

The reviewed literature highlights significant progress in incident reporting systems across various domains, yet notable gaps persist. Traditional frameworks such as CIRS in healthcare emphasize structured reporting and data analysis but rely on centralized databases, making them prone to data manipulation and limited transparency. Surveys on existing systems reveal challenges like lack of anonymity, incomplete reporting, and weak data security. While Blockchain-based approaches, such as decentralized urban incident reporting and water monitoring solutions, address transparency and immutability using smart contracts and IPFS, they often remain domain-specific and lack unified architectures that combine geolocation-based real-time reporting, scalability, and incentive mechanisms. Hence, there exists a clear research gap for a comprehensive, decentralized, and citizen-centric incident reporting framework that ensures transparency, security, traceability, and active community participation.

Table 2.1: Comparison of key related works in decentralized and incident reporting systems.

Paper Name	**Key Concept**	**Performance**	**Strengths**	**Limitations**
Decentralized Incident Reporting: Mobilizing Urban Communities with Blockchain [1]	Blockchain-based urban incident reporting using smart contracts and IPFS for transparency, immutability, and real-time tracking.	Achieved secure, tamper-proof, and verifiable data reporting with low latency using Quorum Blockchain.	Transparent, decentralized, and scalable system enabling real-time visualization and citizen participation.	Limited to permissioned environments; lacks AI-based verification or prioritization of incidents.
Critical Incident Reporting System (CIRS) for Healthcare [2]	Centralized framework for reporting and analyzing medical incidents to improve patient safety and risk management.	Enhanced patient safety through structured reporting and trend-based risk mitigation.	Structured interface, audit trail, and continuous quality improvement mechanisms.	Centralized model limits transparency and real-time sharing across institutions.
Deployment of CIRS in Public Styrian Hospitals: A Five-Year Perspective [3]	Practical deployment of a ealthcare CIRS emphasizing non-punitive culture and data-driven safety improvements.	Improved reporting rates and timely intervention across multiple hospital departments.	Promotes open error-reporting culture; effective cross-departmental risk analysis.	Manual verification and limited automation; potential delays in large-scale data aggregation.
A Survey on Incident Reporting and Management Systems [4]	Comparative study of existing reporting systems focusing on usability, reliability, anonymity, and data security.	Identified strengths of leading systems (e.g., Ushahidi) and gaps in anonymity and data reliability.	Comprehensive evaluation across multiple platforms; highlights best practices and deficiencies.	Lack of Blockchain-based systems; suggests need for privacy-preserving and tamper-proof solutions.
A BC Approach for Migrating a Cyber-Physical Water Monitoring Solution to a Decentralized Architecture [5]	Hybrid Blockchain framework integrating ERC-20 and ERC-1155 tokens for transparent rewards and subscription discounts.	Improved transparency, engagement, and trust between water providers and users.	Tokenization of rewards, automated discount management, and decentralized storage with IPFS.	Complex smart contract management; limited scalability and interoperability testing.

2.4 Summary of Chapter

This chapter reviewed existing literature on incident reporting systems across various sectors, emphasizing the evolution from traditional centralized models to emerging decentralized, Blockchain-based frameworks. It examined key studies on urban governance, healthcare safety, and environmental monitoring, analyzing their objectives, methodologies, and limitations. While conventional systems like CIRS enhance structured reporting and risk management, they suffer from issues of data manipulation, limited transparency, and low public trust. In contrast, Blockchain-based approaches leverage smart contracts, IPFS, and tokenization to achieve transparency, immutability, and citizen engagement. However, these solutions remain domain-specific and lack a unified, scalable, and geolocation-enabled architecture. The review thus identifies the need for an integrated, decentralized framework that ensures secure, tamper-proof, and community-driven incident reporting to enhance accountability and urban governance.

Chapter 3

System Design

3.1 Introduction

This chapter describes the overall system design of the proposed Blockchain-based incident reporting system. It explains the system architecture, key design diagrams, and major modules that enable secure and transparent incident reporting. The chapter also outlines the algorithms used, software and hardware requirements, database design, project timeline, and user interface layout, providing a clear blueprint for the implementation of the system.

3.2 Overall Architecture

Architecture Overview

The system follows a three-layered architecture separating presentation, business logic, and data storage to ensure scalability, modularity, and easier maintenance. The frontend manages user interaction, the backend processes application logic, and the data layer handles decentralized and structured storage.

Presentation Layer

- **Web UI:** Main interface through which users interact with the system.
- **User Interface:** Allows users to report new incidents and view existing reports.
- **Verifier Interface:** Dedicated interface for authorized users to verify reported incidents.
- **Map Interface:** Displays incident locations geographically using an interactive map.

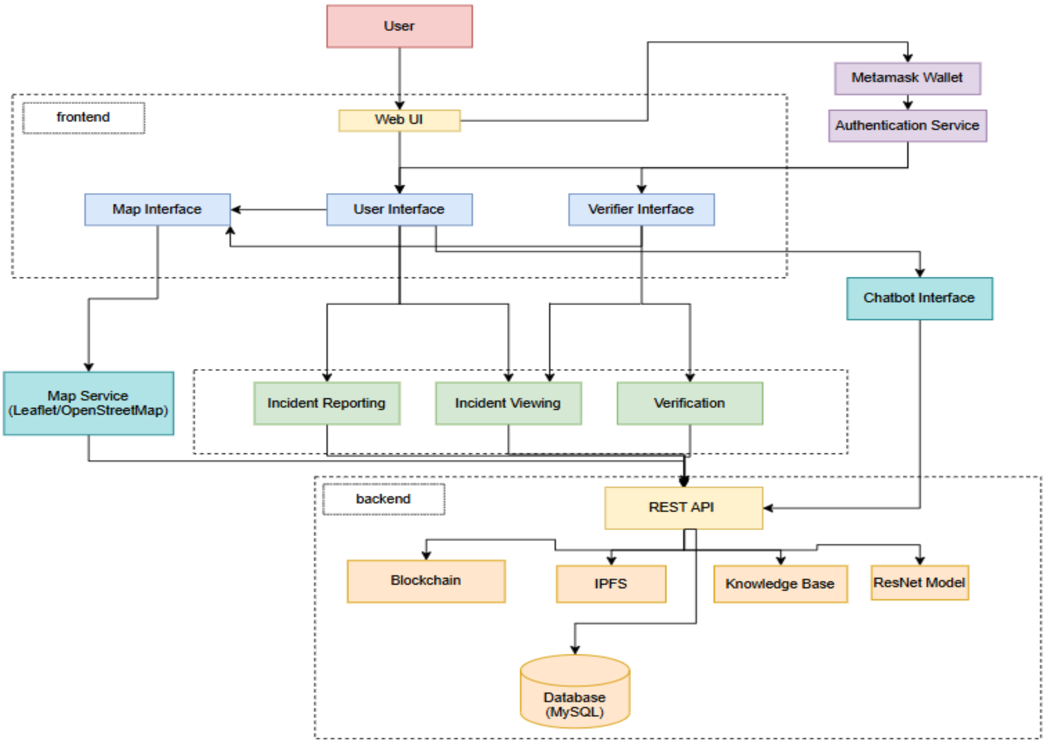

Figure 3.1: System Architecture.

- **MetaMask Wallet & Authentication Service:** Provides decentralized authentication and verifies user identity before accessing system features.
- **Chatbot Interface:** Assists users by answering queries related to incidents and system usage.

Business Logic Layer

- **Incident Reporting:** Handles incident submission and processes form data before storing it in the system.
- **Incident Viewing:** Retrieves incident data from backend services and presents it to users.
- **Verification:** Allows verifiers to review reports and update incident verification status.
- **Map Service (Leaflet/OpenStreetMap):** Processes geographic coordinates and displays incident markers on the map.

- **REST API/Backend:** Manages communication between frontend interfaces and backend services.

Data Layer

- **Blockchain:** Ethereum-based ledger used to store critical incident records ensuring transparency and immutability.
- **IPFS:** Decentralized storage system used to store images or media evidence associated with incidents.
- **Knowledge Base:** Stores structured information and responses used by the chatbot system.
- **ResNet Model:** Machine learning model used for image analysis and classification of incident-related images.
- **Database (MySQL):** Stores structured metadata such as incident details, verification status, and system records.

Data Flow

1. User authenticates using the MetaMask Wallet.
2. Web UI directs the user to the appropriate interface (User or Verifier).
3. User actions such as reporting or viewing incidents are sent to the REST API.
4. Backend processes the request and stores data across Blockchain, IPFS, and MySQL database.
5. Map Service retrieves incident locations and displays markers on the interactive map.
6. Chatbot Interface accesses the Knowledge Base and backend services to respond to user queries.

3.3 Design Diagrams

3.3.1 Sequence Diagram

Purpose:
Illustrates the order of interactions among system components during the three primary workflows: reporting, viewing, and verifying incidents.

Report Incident Flow:

- **User** logs in via MetaMask → **User Interface** displays dashboard.
- User selects "Report Incident" and fills form → **Incident Service** processes submission.
- **Database** stores incident data → **Blockchain System** creates new block.
- **Map Service** adds yellow marker → Success confirmation returned.

View Incident List Flow:

- **User** selects "View Incident List" → **Incident Service** fetches data from **Database**.
- **Map Service** generates map with color-coded markers.
- User clicks incident → Detailed information displayed.

Verify Incident Flow:

- **Verifier** logs in and selects incident → **Incident Service** processes verification.
- **Database** and **Blockchain System** update status → Marker changes to green.

Flow Summary:
Authentication → Action Selection → Service Processing → Database & Blockchain Operations → Response Delivery.

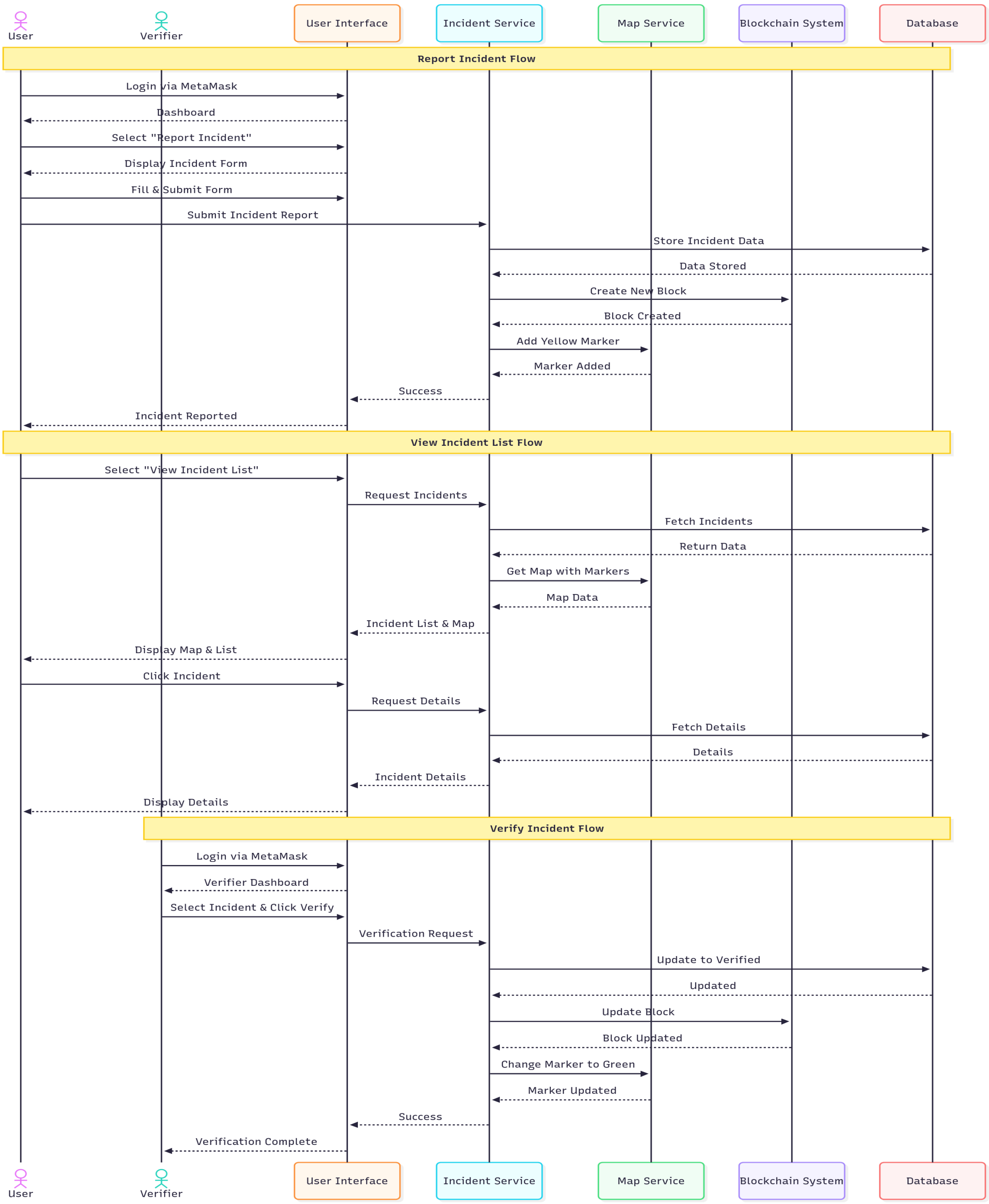

Figure 3.2: Sequence Diagram

3.3.2 Use Case Diagram

Purpose:

Depicts the functional interactions between actors (User and Verifier) and core system use cases.

Actors:

- **User:** Reports and views incidents.
- **Verifier:** Verifies incidents in addition to viewing them.

Core Use Cases:

- **Login:** MetaMask wallet authentication for both actors.
- **View Incidents:** Includes View Map and View List functionalities.
- **Report Incident:** User selects category and fills form with incident details.
- **Verify Incident:** Verifier approves/rejects incidents to update status.

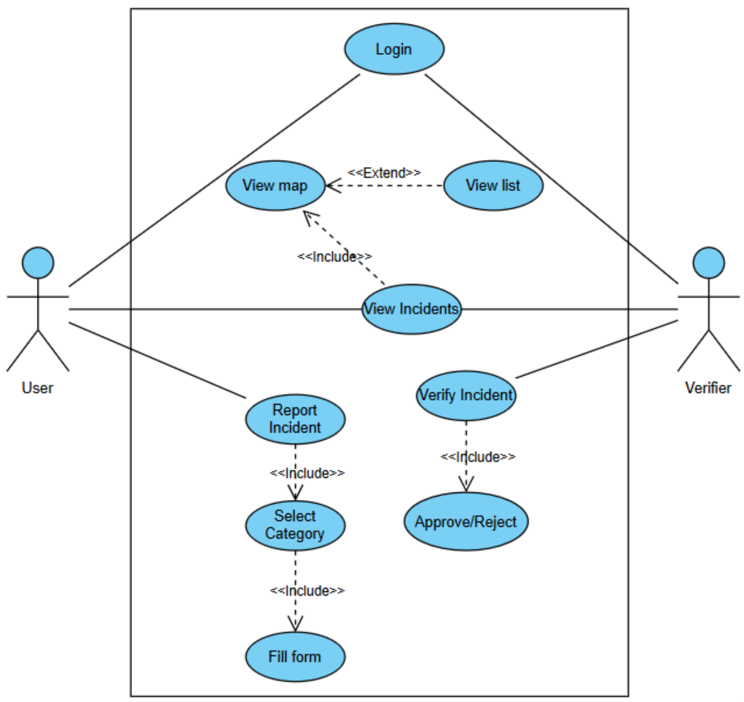

Figure 3.3: Use Case Diagram

3.4 Summary of Chapter

This chapter presents the system design of the proposed blockchain-based incident reporting platform. It describes the overall architecture of the system and explains how different components are organized. Design diagrams and architectural representations are used to illustrate the interaction between the user interface, backend services, decentralized storage, and the blockchain network. These diagrams provide a clear understanding of the system workflow, data flow, and component relationships involved in the reporting, verification, and visualization of incidents.

Chapter 4

System Implementation

4.1 Introduction

This chapter presents the implementation of the proposed blockchain-based incident reporting system. It describes how the conceptual design of the system is translated into a functional application through the development of different system modules. The chapter begins with an explanation of the main modules of the system and their roles in enabling incident reporting, verification, and visualization. It further discusses the development approach adopted for building the system, along with the tools and technologies used during implementation. The chapter also outlines the system requirements necessary for running the application and provides an overview of the work distribution among team members involved in the project development.

4.2 Module Division

4.2.1 Incident Reporting Module

- **MetaMask Wallet Authentication:** Allows users to connect their MetaMask wallet for decentralized login without traditional usernames or passwords.
- **Incident Submission Interface:** Enables users to report incidents by providing description, location coordinates, timestamp, and image evidence.
- **Unique Incident ID Generation:** Automatically generates a unique identifier for each incident to support tracking and referencing.
- **Input Data Validation:** Ensures that essential fields such as description, location, and timestamp are correctly provided before submission.

- **Blockchain Transaction Trigger:** Sends the validated incident data to the smart contract for secure and permanent recording.

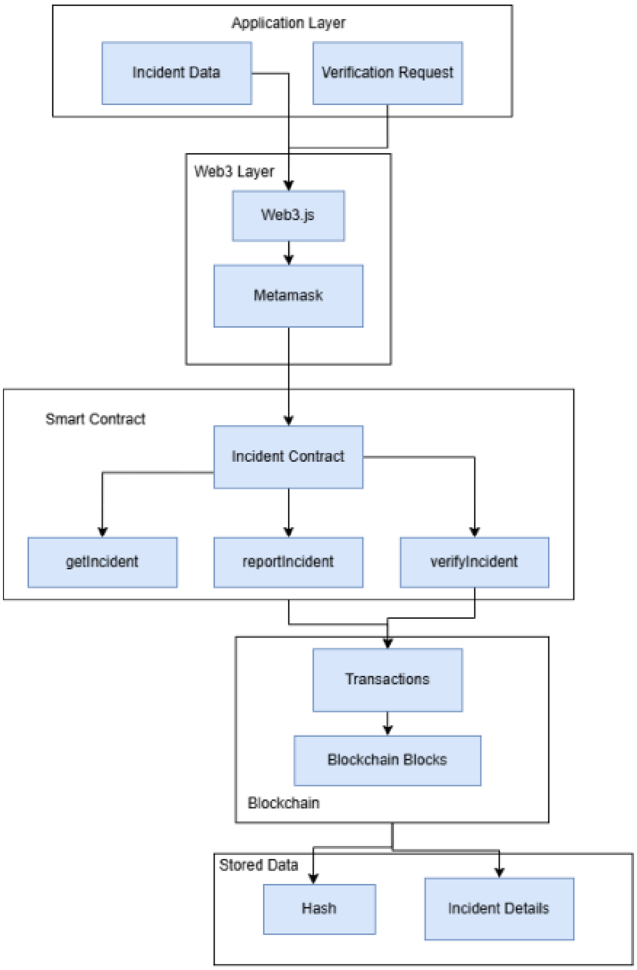

Figure 4.1: Incident Reporting Module

4.2.2 Blockchain Module

- **Smart Contract Deployment:** Implements Ethereum smart contracts to manage incident recording and retrieval.

- **Immutable Data Storage:** Ensures that once an incident is recorded on the Blockchain it cannot be altered or deleted.

- **Transaction Processing:** Handles Blockchain transactions generated when incidents are reported or verified.

- **Incident Data Retrieval:** Enables the application to fetch incident details stored on the Blockchain.

- **Decentralized Transparency:** Maintains a tamper-proof and publicly verifiable record of all incidents.

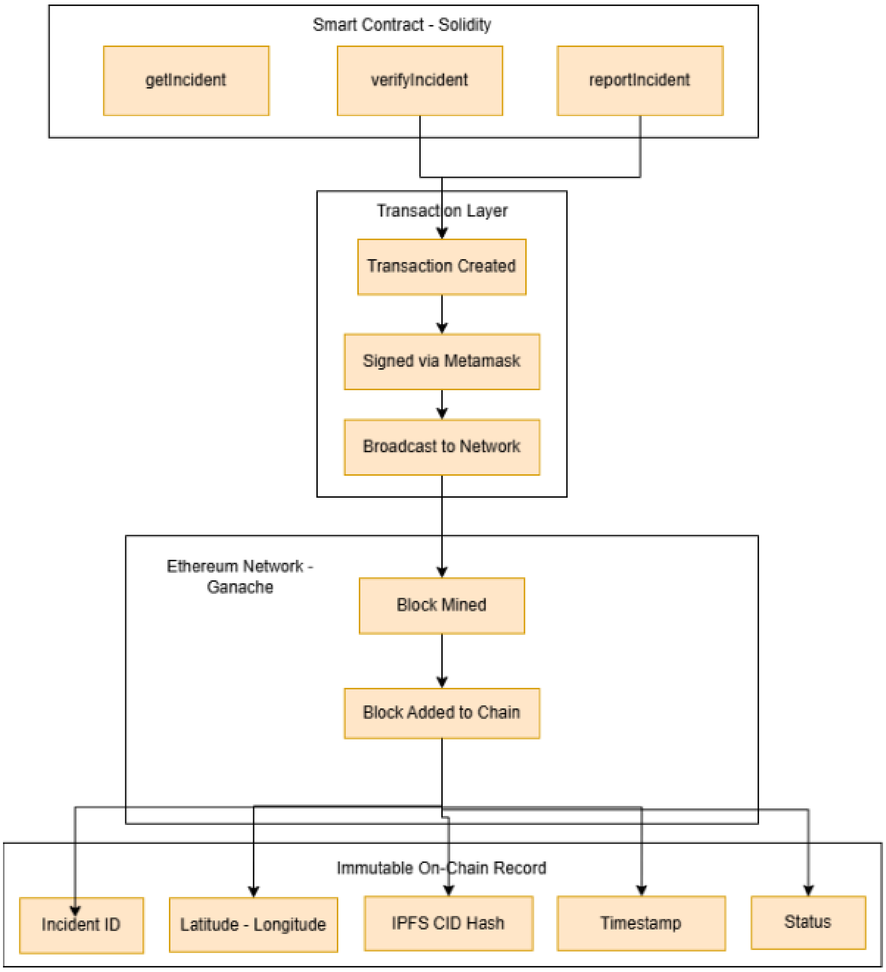

Figure 4.2: Blockchain Module

4.2.3 Map-Based Visualization Module

- **Interactive Map Interface:** Displays incidents on a digital map for intuitive geographic visualization.

- **Location Marker Placement:** Automatically places markers on the map based on the reported incident coordinates.

- **Incident Detail Popups:** Shows relevant incident information when users select a map marker.
- **Real-Time Map Updates:** Updates the map dynamically when new incidents are recorded.
- **Geographic Insight:** Helps users easily identify affected areas and incident clusters.

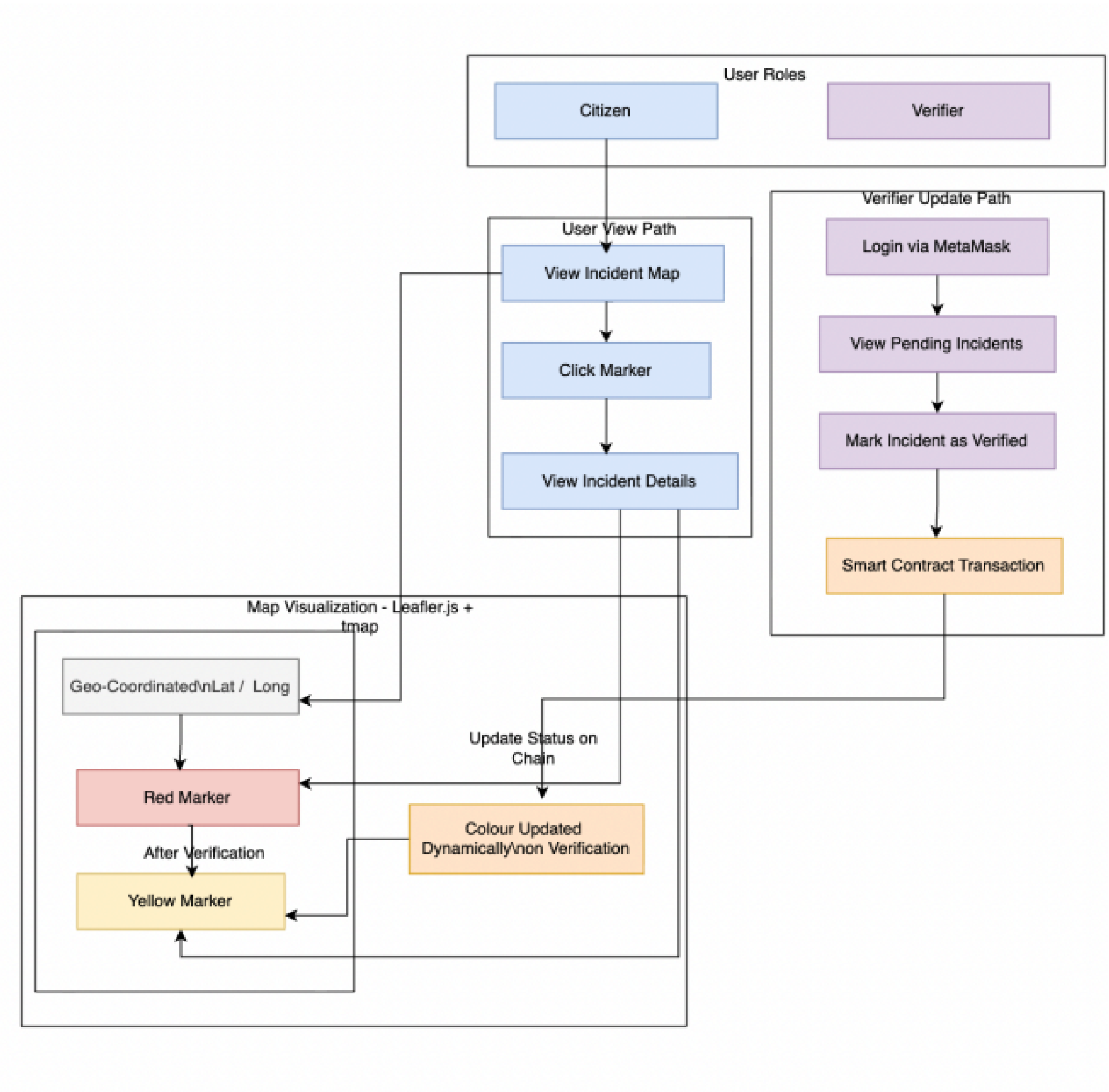

Figure 4.3: Map-based Visualisation Module

4.2.4 Off-Chain Storage Module

- **IPFS File Storage:** Stores images and evidence files in a decentralized manner using IPFS, generating a unique Content Identifier (CID) for secure access.
- **Blockchain Linking:** Associates the generated CID with the corresponding incident by storing it in the smart contract for verification and retrieval.
- **MySQL Description Storage:** Stores only the incident description mapped with the transaction hash (Tx Hash) as a reference key.
- **Data Retrieval Mechanism:** Fetches the description from the database using the Tx Hash obtained from the blockchain.

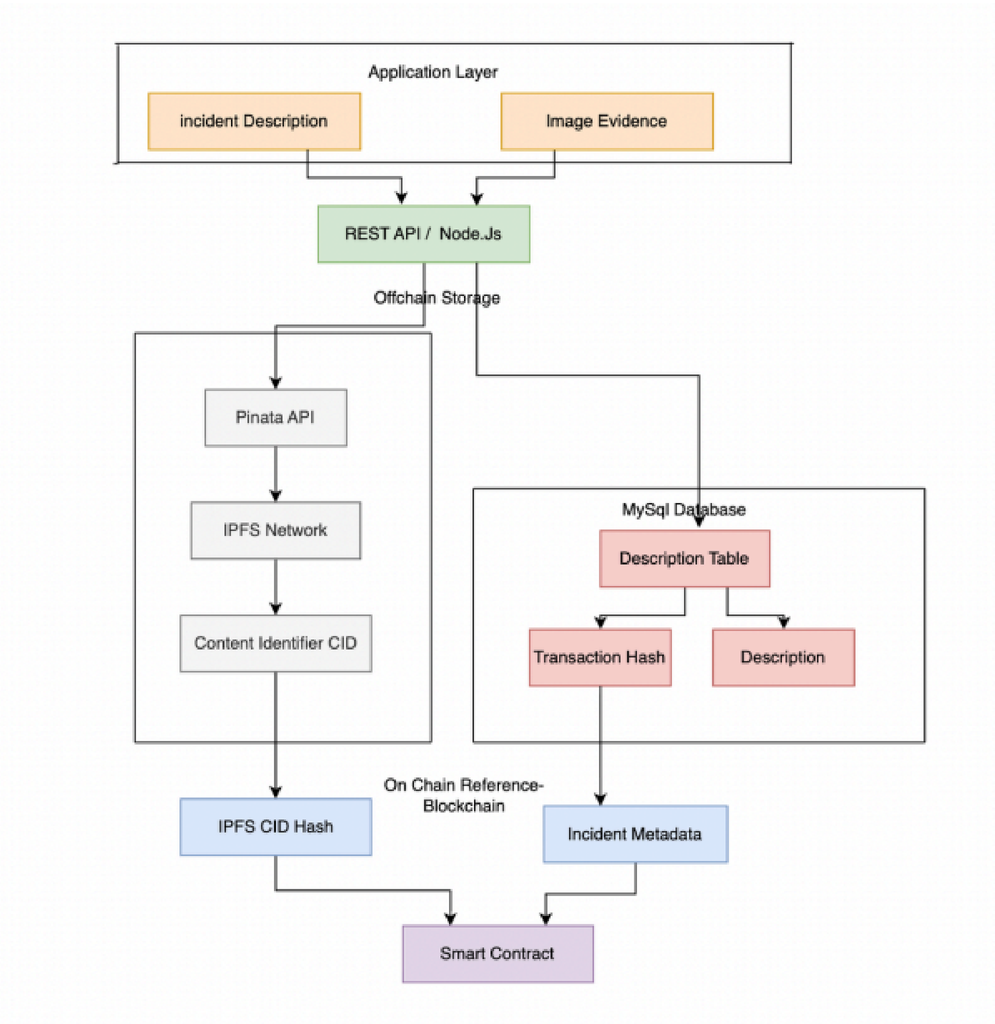

Figure 4.4: Off-chain Storage

4.2.5 Incident Verification Module

- **Verification Decision Recording:** Enables verifiers to approve or reject reported incidents.
- **Blockchain Status Update:** Records verification outcomes on the Blockchain to maintain transparency.
- **Transparent Verification Tracking:** Maintains a clear record of verification actions associated with each incident.

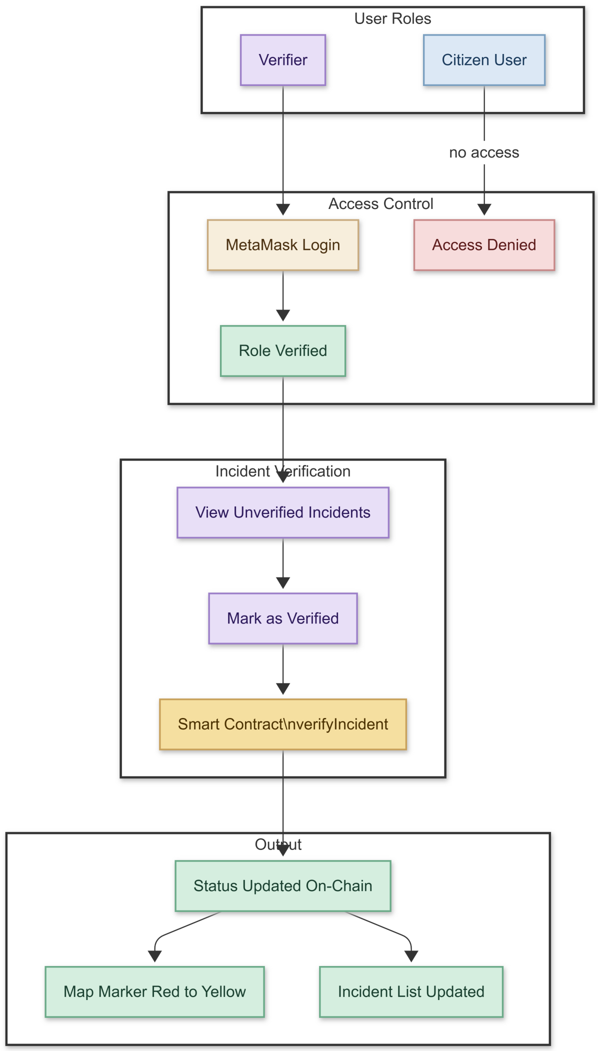

Figure 4.5: Verification Module

4.2.6 Chatbot Module

- **Automated User Assistance:** Provides guidance to users on how to report incidents and navigate the application.
- **Query Handling:** Responds to common user questions related to system features and reporting procedures.
- **Navigation Support:** Assists users in locating key sections such as reporting forms and the incident map.
- **Information Retrieval:** Supplies quick explanations about the reporting process and verification workflow.
- **Enhanced User Experience:** Improves accessibility and usability by offering instant assistance.

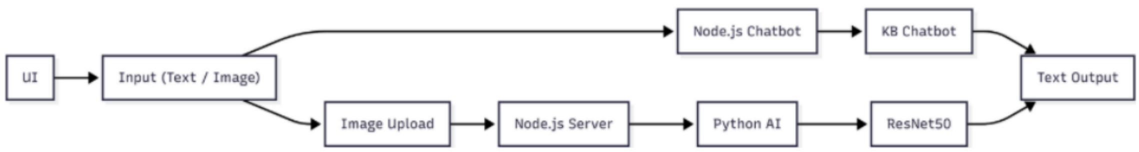

Figure 4.6: Chatbot Module

4.3 Tools and Technologies

4.3.1 Hardware Requirements

- **Processor:** Intel Core i3 or equivalent (i5/i7 recommended)
- **RAM:** 4 GB minimum, 8 GB recommended
- **Storage:** 10 GB free disk space
- **Network:** Stable internet connection
- **Device:** Desktop/Laptop

4.3.2 Software Requirements

- Operating System: Windows 10/11, macOS 10.14+, or Linux (Ubuntu 20.04+)
- Solidity (v0.8.19)
- Truffle (v5.11.5)
- Ganache (v7.9.1)
- Web3.js (v1.10.0)
- MetaMask
- IPFS
- Leaflet (Map API)
- HTML / CSS / JavaScript
- Web Browser

4.4 Work breakdown and Responsibilities

- **Susen Ann George** – Designed and implemented smart contracts in Solidity and developed the Blockchain logic for incident recording and verification. Trained the chatbot to respond to user queries and worked on image-based incident description and precaution suggestions.
- **Siona Kurisinkal Biju** – Handled incident creation, on-chain data storage, and verification logic. Implemented map-based visualization using Leaflet and OpenStreetMap. Managed off-chain storage for incident descriptions and integrated IPFS for image uploads and retrieval. Performed final integration and system testing.
- **Treesa Maria Cinil** – Implemented map-based visualization using Leaflet and OpenStreetMap. Developed the incident verification mechanism.
- **Niya Sony** – Designed and implemented the web user interface using HTML and CSS. Contributed to the preparation and design of project presentations.

- **All members** – Collaborated on literature surveys, feasibility analysis, code reviews, progress meetings, and presentation preparations to ensure cohesive and successful project execution.

4.5 Key Deliverables

Decentralized Incident Reporting Platform

- Provides a web-based interface that enables citizens to report urban incidents such as accidents, hazards, or infrastructure issues in real time.
- Allows users to submit incident details including description, geographic location, timestamp, and image evidence through a simple reporting form.
- Integrates MetaMask wallet authentication to identify users through their Ethereum wallet address without relying on traditional login credentials.

Blockchain-Based Incident Management

- Implements Ethereum smart contracts to securely record incident details and maintain a transparent record of all submitted reports.
- Ensures immutability by storing critical incident data on the Blockchain so that it cannot be altered or deleted after submission.
- Supports incident verification by enabling authorized users to review reports and update their verification status through Blockchain transactions.

Interactive Map Visualization

- Displays all reported incidents through an interactive map interface developed using Leaflet and OpenStreetMap.
- Places location markers based on the geographic coordinates provided during incident reporting.
- Allows users to click on markers to view detailed incident information such as description, reporting time, and verification status.

Decentralized Image Storage

- Stores uploaded incident images using the InterPlanetary File System (IPFS) to avoid storing large files directly on the Blockchain.
- Generates a unique content identifier (CID) for each uploaded image to enable secure and decentralized file referencing.
- Links the generated CID with the corresponding incident record on the Blockchain to allow retrieval of image evidence when viewing incident details.

Chatbot Assistance System

- Provides an automated chatbot interface that assists users in navigating the application and understanding the incident reporting process.
- Responds to common user queries related to system usage, reporting steps, and platform features.
- Offers precautionary suggestions and guidance based on the type of incident described by the user.

4.6 Project Timeline

Figure 4.7: Work Plan of the Proposed System

4.7 Summary of Chapter

This chapter described the implementation of the proposed blockchain-based incident reporting system. It explained the different modules of the system and their roles in enabling incident reporting, verification, and visualization. The chapter also discussed the development approach adopted for building the system, along with the tools and technologies used during implementation. In addition, it outlined the system requirements necessary for deployment and presented the distribution of work among the team members involved in the project development.

Chapter 5

Results and Discussions

5.1 Overview

The proposed decentralized incident reporting system was evaluated through a series of experimental scenarios to analyze its functionality, performance, and reliability. The evaluation considered multiple categories of incidents, including transportation issues, infrastructure damage, water and sanitation problems, and safety or security incidents.

The testing process focused on key system performance metrics such as IPFS upload time, Blockchain transaction execution time, and the total time required for successful incident submission. These metrics help assess the efficiency of decentralized storage and blockchain integration within the system.

The experiments were conducted in a controlled environment using the implemented prototype. The obtained results demonstrate that the system maintains stable performance across different incident categories while ensuring transparency, data integrity, and secure storage of incident records.

5.2 Testing

The implemented system interfaces were tested to verify their functionality and usability. The following screenshots illustrate the major components of the system, including user authentication, incident reporting, Blockchain block creation, incident verification, map visualization, and chatbot assistance.

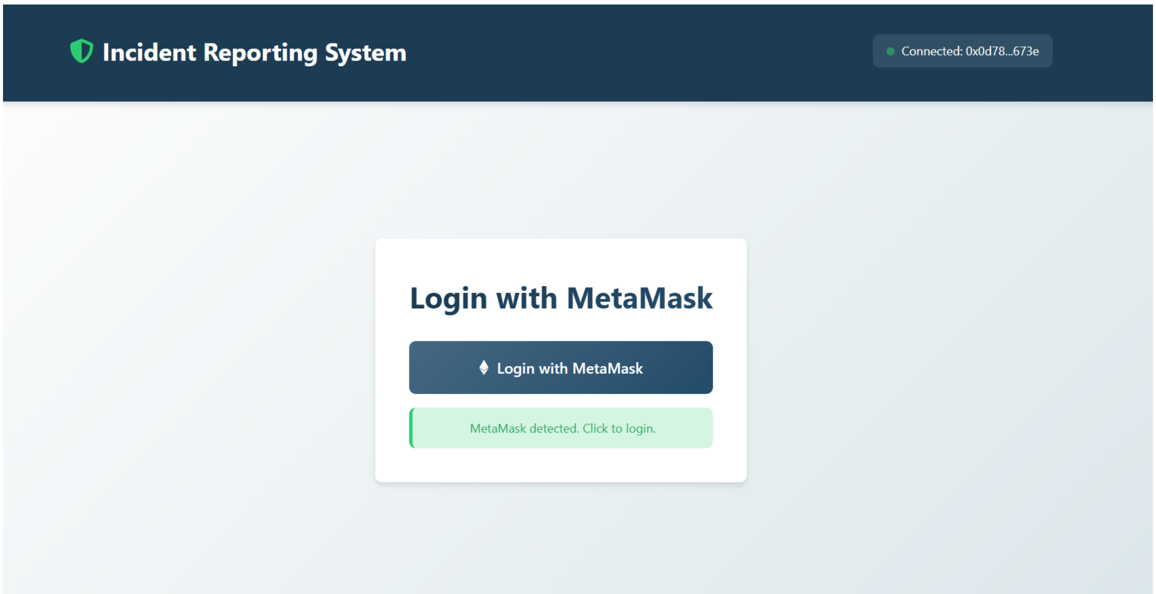

Figure 5.1: Login Page

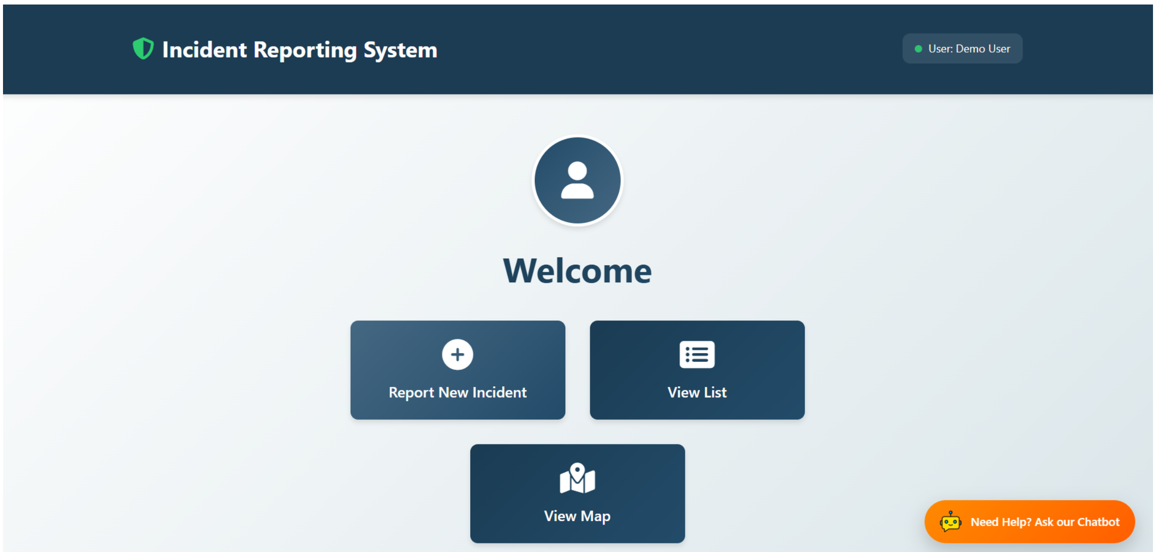

Figure 5.2: User Page

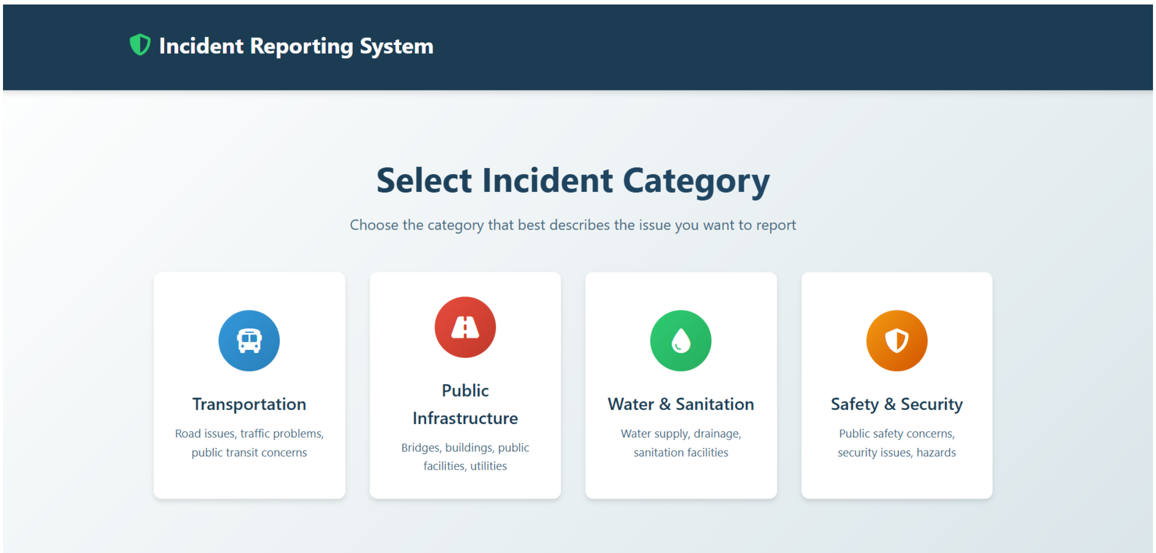

Figure 5.3: Incident Category Selection

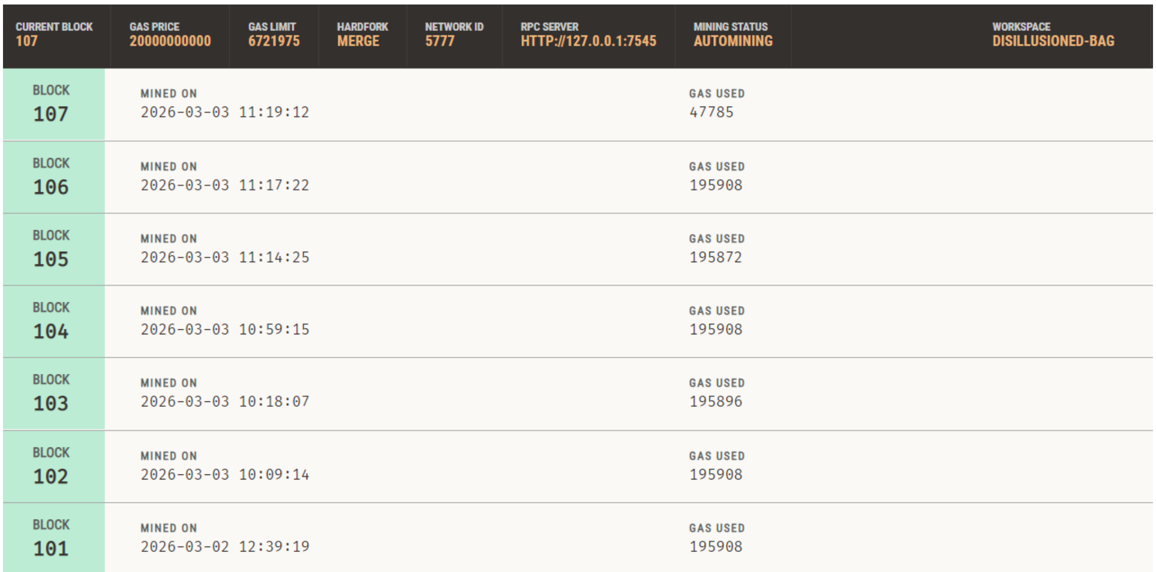

Figure 5.4: Blockchain Blocks Created in Ganache

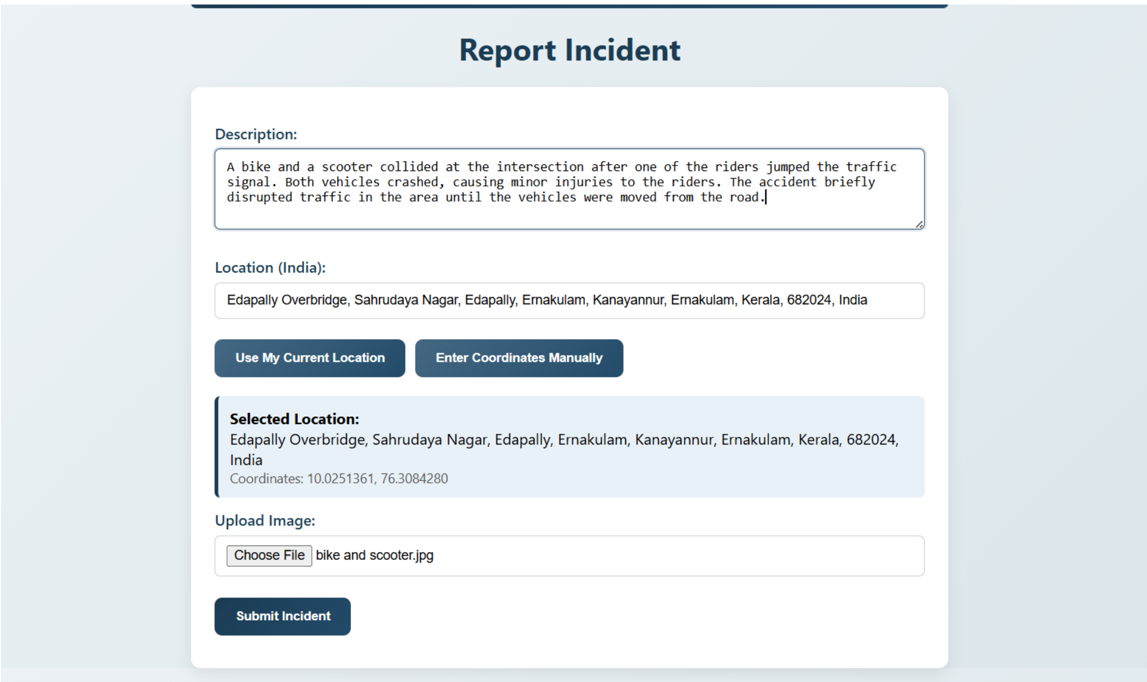

Figure 5.5: Incident Reporting Page

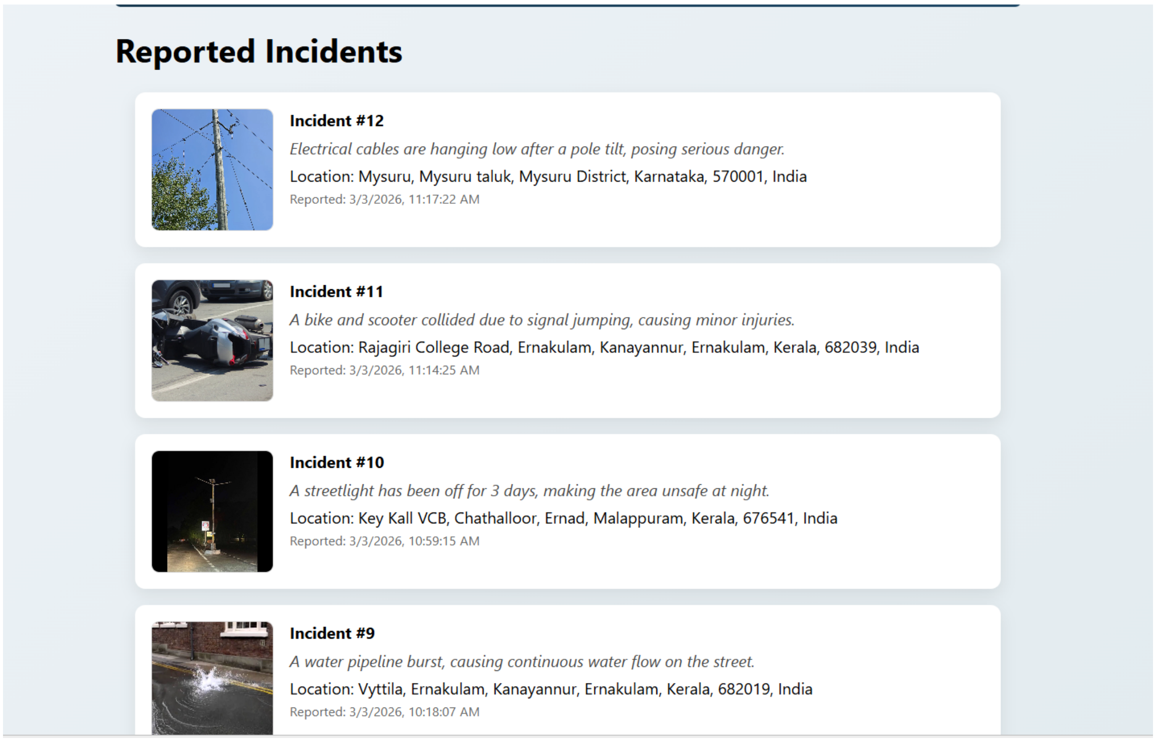

Figure 5.6: List of Reported Incidents

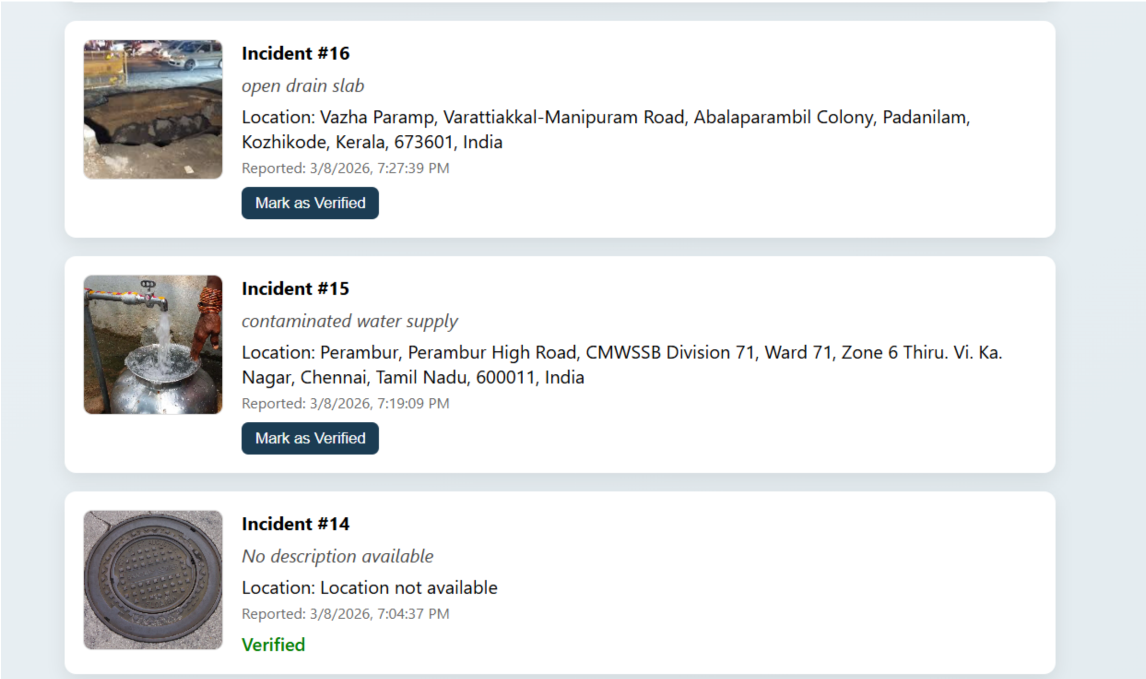

Figure 5.7: Incident Verification Page

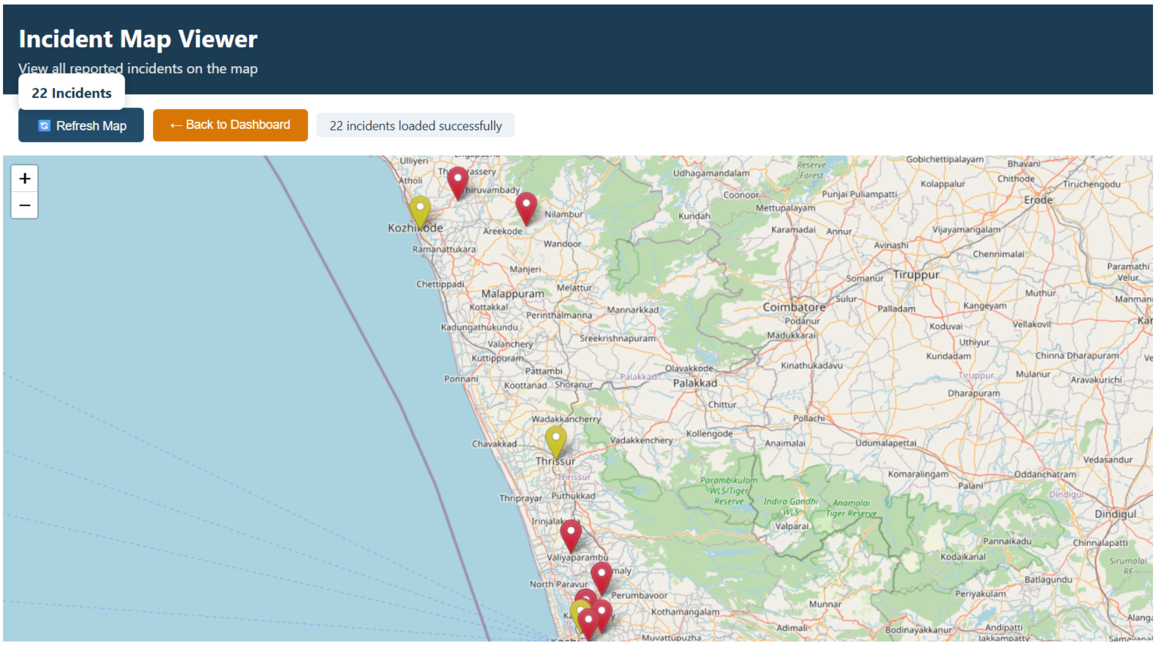

Figure 5.8: Map Visualization of Reported Incidents

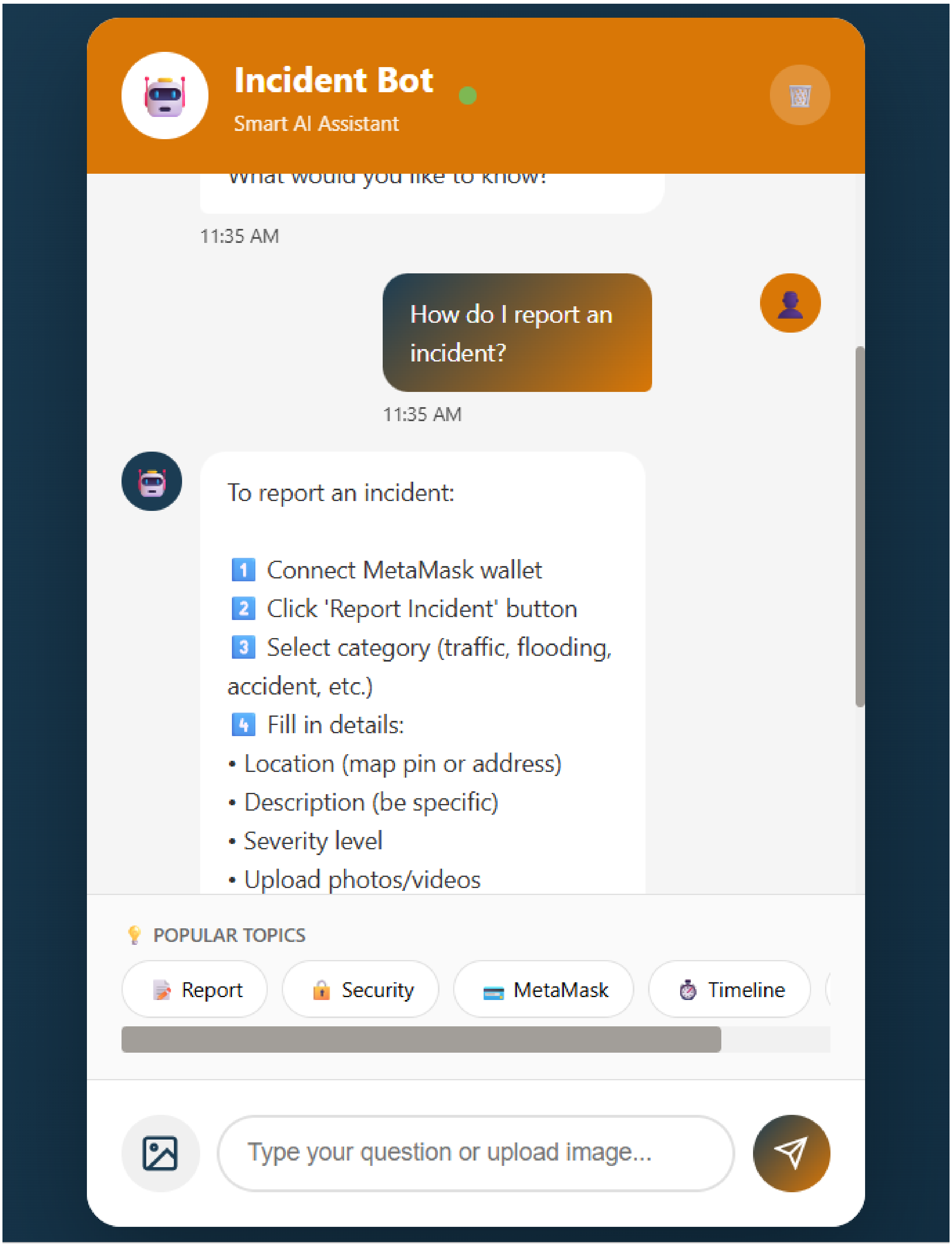

Figure 5.9: AI-assisted Chatbot Interface

5.3 Quantitative Analysis

5.3.1 Efficiency Metrics

To evaluate system performance, several experiments were conducted using different categories of incidents. The primary metrics measured during testing include IPFS upload time, Blockchain transaction execution time, and the total incident submission time.

Table 5.1 presents the performance measurements obtained during testing.

Table 5.1: Performance Metrics for Different Incident Categories

Incident Category	**IPFS Upload Time (s)**	**Blockchain Time (s)**	**Total Submission Time (s)**
Transportation	2.508	4.142	8.401
Infrastructure	3.186	4.665	9.453
Water and Sanitation	1.733	2.828	6.248
Safety and Security	3.782	2.961	8.584
Average	2.802	3.649	8.172

The results indicate that the system maintains consistent response times across different incident categories. The integration of IPFS enables efficient decentralized storage of multimedia evidence, while Blockchain ensures secure and immutable recording of incident information. The average incident submission time remains within a reasonable range, demonstrating the practical feasibility of the proposed decentralized approach.

5.4 Discussion

The experimental evaluation shows that the proposed system performs reliably while integrating Blockchain and decentralized storage technologies. The use of IPFS significantly reduces the amount of data stored directly on the Blockchain, thereby improving efficiency and scalability.

Although minor variations in submission time were observed among different incident categories, the overall system response remained stable. The implemented interfaces, including the incident reporting form, map-based visualization, and AI-assisted chatbot, further improve usability by enabling users to easily submit incidents, view reported locations, and obtain guidance during the reporting process.

These features collectively contribute to an effective platform for transparent and secure incident reporting.

5.5 Summary of Chapter

This chapter presented the experimental evaluation of the proposed decentralized incident reporting system. The system was tested using multiple incident categories to assess its performance and operational reliability. Performance metrics such as IPFS upload time, Blockchain transaction execution time, and total incident submission time were analyzed using experimental measurements.

The results demonstrate that the system provides consistent performance while ensuring transparency, data integrity, and secure storage through Blockchain and IPFS integration. Additionally, the implemented interfaces enhance usability and improve the overall reporting experience for users. The findings confirm that the proposed system is a practical and efficient solution for decentralized incident reporting and urban incident management.

Chapter 6

Conclusion & Future Scope

6.1 Conclusion

The proposed decentralized incident reporting system provides a reliable and transparent platform for reporting and managing urban incidents. By integrating Blockchain technology with IPFS-based decentralized storage, the system ensures secure and immutable storage of incident records while efficiently handling multimedia evidence. The use of smart contracts enables automated incident registration and verification, while role-based access control ensures that only authorized users can validate incidents.

The system also incorporates map-based visualization to display incidents according to their geographic locations, improving situational awareness and enabling users to easily identify affected areas. In addition, the AI-assisted chatbot enhances user interaction by guiding users during the reporting process and answering queries related to the system.

Experimental evaluation shows that the system performs efficiently with consistent response times during incident submission and verification. The results confirm that the proposed architecture effectively combines Blockchain transparency, decentralized storage, and intelligent user assistance to support reliable incident management. Overall, the system demonstrates strong potential for application in smart city environments by encouraging citizen participation and improving the efficiency of urban incident reporting.

6.2 Future Scope

- **Integration with Public Blockchain** – Connecting the system to public Blockchain networks to improve decentralization and accessibility.
- **Advanced Machine Learning for Incident Detection** – Implementing ML models to automatically classify incidents and estimate their severity from uploaded

images.

- **Integration with Emergency Response Systems** – Linking the platform with government or municipal emergency services for faster incident response.

- **Scalability Enhancement** – Optimizing Blockchain transactions and system architecture to support a larger number of users and reports.

- **Mobile Application Development** – Developing a dedicated mobile application to make incident reporting more convenient and accessible for users.

6.3 Summary of Chapter

This chapter presented the conclusion of the the proposed decentralized Blockchain-based incident reporting system. The key outcomes of the system, including improved transparency, secure data storage, and efficient incident management, were discussed. The chapter also highlighted how the integration of Blockchain, IPFS, map-based visualization, and AI assistance enhances the reliability and usability of the system. In addition, potential future improvements such as integration with public Blockchain networks, machine learning-based incident analysis, and connection with emergency response systems were outlined to further enhance the system's capabilities.

References

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Appendix A: Presentation

BLOCKCHAIN BASED INCIDENT REPORTING SYSTEM

FINAL PRESENTATION

PROJECT GUIDE:

Ms.Jisha Mary Jose

GROUP MEMBERS:

Susen Ann George (U2203202)
Siona Kurisinkal Biju (U2203194)
Treesa Maria Cinil (U2203205)
Niya Sony (U2203167)

10-03-2026

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INTRODUCTION

- Urban areas frequently face issues such as accidents, flooding and safety hazards.
- These incidents can disrupt daily life, cause delays, and create risks for public safety.
- Rapid urbanization and increasing population make it more challenging to monitor and manage such incidents effectively.
- Timely reporting and proper coordination are essential to ensure quicker response and improved urban safety.

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PROBLEM DEFINITION

Current incident reporting systems are centralized and lack transparency, leading to delays, data manipulation risks, and reduced public trust. There is no unified platform for real-time, verifiable, and tamper-proof reporting.

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PROJECT OBJECTIVE

- To develop a decentralized incident reporting system using Blockchain technology that overcomes the limitations of centralized platforms.
- To enhance citizen participation in reporting public safety and urban issues.
- To implement an Ethereum based backend with smart contracts for secure report submission and storage.
- To improve urban management through real-time incident data for better response and planning.

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PURPOSE AND NEED

Purpose of the Project

- Develop a decentralized Blockchain-based incident reporting system for urban areas.
- Support efficient urban management by authorities through real-time data inputs.

Need for the Project

- Limited citizen engagement in incident reporting reduces effective public safety monitoring.
- Urban authorities require reliable and timely data for improved decision-making and resource allocation.

10-03-2026

6

LITERATURE SURVEY

PAPER 1 - Petschnig, W., & Haslinger-Baumann, E. (2017). Critical Incident Reporting System (CIRS): a fundamental component of risk management in health care systems to enhance patient safety. Safety in Health, 3(1), 9.

Objective

To evaluate the effectiveness of Critical Incident Reporting System (CIRS) in improving patient safety and risk management.

Method

- Literature review using CINAHL, PubMed, MEDLINE, Cochrane Library, Google Scholar, and Thieme e-book library.
- Keywords: CIRS, patient safety, incident reporting, near miss, voluntary reporting systems.
- Studies and reports analyzed for incident types and reporting patterns.

Key Findings

- CIRS improves patient safety when supported by management and a no-blame culture.
- Common incidents: medication errors, patient falls, communication failures.
- Nurses report most incidents, while doctors report less frequently.

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Paper 2 - Sendlhofer, G., Schweppe, P., Sprincnik, U., Gombotz, V., Leitgeb, K., Tiefenbacher, P., ... & Brunner, G. (2019). Deployment of Critical Incident Reporting System (CIRS) in public Styrian hospitals: a five year perspective. BMC health services research, 19(1), 412.

Objective

To assess five years of CIRS use in hospitals to evaluate reporting behavior and identify areas for improvement.

Method

- Retrospective observational study across 22 regional and 1 university hospital in Austria.
- Voluntary CIRS reporting using R2C software with predefined and optional fields.
- Reports were anonymized and reviewed by expert teams (physicians, nurses, risk managers, jurists).
- Pseudonymized reports (2013–2017) analyzed for frequency and categories using R version 3.4.2.

Key Findings

- 2,504 incidents reported over 5 years with a steady annual increase.
- Most reports from surgery, anesthesiology, and intensive care units.
- Common errors: patient identification, medication management, device handling, and communication failures.

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Paper 3 - Qureshi, A., Megias, D., & Rifā-Pous, H. (2018). A survey on incident reporting and management systems.

Objective

To analyze and compare incident reporting systems based on features, usability, performance, security, and anonymity.

Method

- Surveyed existing web and mobile incident reporting systems and analyzed features such as usability, GPS tracking, reliability, security, and anonymity.
- Compared systems as public (non-anonymous) and private (anonymous) using user ratings, documentation, and experimental results.

Key Findings

- Ushahidi performs best among public systems with strong usability, reliability, and performance.
- Most systems are free and GPS-enabled, but many lack strong security, reliability, and anonymous reporting features.

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Paper 4 - Pahontu, B. I., Petcu, A., Predescu, A., Arsene, D. A., & Mocanu, M. (2023). A blockchain approach for migrating a cyber-physical water monitoring solution to a decentralized architecture. Water, 15(16), 2874.

Objective

To migrate a water monitoring and incident reporting system from a centralized architecture to a blockchain-based decentralized model.

Method

- Implemented on Ethereum using smart contracts.
- Used ERC-20 tokens for reporting rewards and ERC-1155 tokens for subscription discounts.
- Integrated crowdsensing and serious gaming to encourage citizen participation.

Key Findings

- Developed a hybrid decentralized system enabling secure reporting, reward distribution, and discount trading.
- Improved trust, transparency, and citizen engagement compared to centralized systems.

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Paper 5 - Putz, B., Vielberth, M., & Pernul, G. (2022, August). BISCUIT-Blockchain Security Incident Reporting based on Human Observations. In Proceedings of the 17th International Conference on Availability, Reliability and Security (pp. 1-6).

Objective

To propose BISCUIT, a decentralized system for transparent and tamper-proof reporting of blockchain security incidents.

Method

- Developed BISCUIT framework with roles: Novice, Expert, and Committee.
- Created a taxonomy for blockchain threats.
- Designed an incident model using binary vectors and similarity matching.

Key Findings

- Human observations are effective: even novice users can report incidents.
- Structured taxonomy improves classification of incidents.
- Blockchain storage ensures tamper-proof and transparent reporting.
- Prototype showed good usability for real incident reporting.

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Paper	Features	Advantages	Limitations
Critical Incident Reporting System (CIRS) (2017)[8]	Records incidents & near misses, voluntary/anonymous, uses conventional DB (no blockchain)	Improves safety, supports openness & learning	Low reporting, incomplete data, no universal standard
Deployment of Critical Incident Reporting System (CIRS) in public Styrian hospitals (2019) [9]	Incident reporting, standardized forms, feedback, no blockchain	Enhances safety, adaptable, supports open culture	Underreporting,incomplete reports, no standard
A survey on incident reporting and management systems (2018) [10]	Multi-channel reporting (app, web, SMS), dashboards, no blockchain	Fast reporting, better citizen–authority coordination	Centralized system, trust issues

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Paper	Features	Advantages	Limitations
A blockchain approach for migrating a cyber-physical water monitoring solution to a decentralized architecture (2023) [11]	Migrates CPS to Blockchain; data stored immutably; smart contracts for reward system	Ensures data integrity & transparency,prevents tampering; improves anomaly detection & response time	Increases system complexity; needs expertise
BISCUIT-Blockchain Security Incident Reporting based on Human Observations (2022). [12]	Community-driven-incident reporting on blockchain; IoT & citizen inputs; smart contracts for categorization & workflows; analytics for patterns	Tamper-proof records; real-time alerts; transparency; public trust & participation	Requires user adoption; blockchain costs (gas fees); secure connectivity needed; depends on input accuracy

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PROPOSED METHOD

- The system is a web-based **Decentralized application built on Ethereum Blockchain** for reliable urban incident management.
- Citizens can report incidents by submitting a **description, geographic location, timestamp, and image evidence** through the application.
- The system is based on a hybrid storage model.
  - Core details (**incident ID, coordinates, timestamp, verification status**) are stored on blockchain using Solidity smart contracts.
  - Detailed descriptions are stored off-chain in a database.
  - Images are stored on IPFS with only the hash stored on-chain to reduce cost.

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- **Smart contracts** manage incident lifecycle, including registration, unique ID generation, and verification updates when authorized verifiers confirm incidents.
- Incidents are visualized on an i**nteractive map using Leaflet and OpenStreetMap**, with color-coded markers for verified/unverified reports and real-time updates.
- An **AI chatbot assists** users by guiding the reporting process, answering system queries, analyzing uploaded images to identify incident type and severity, and suggesting precautionary measures.

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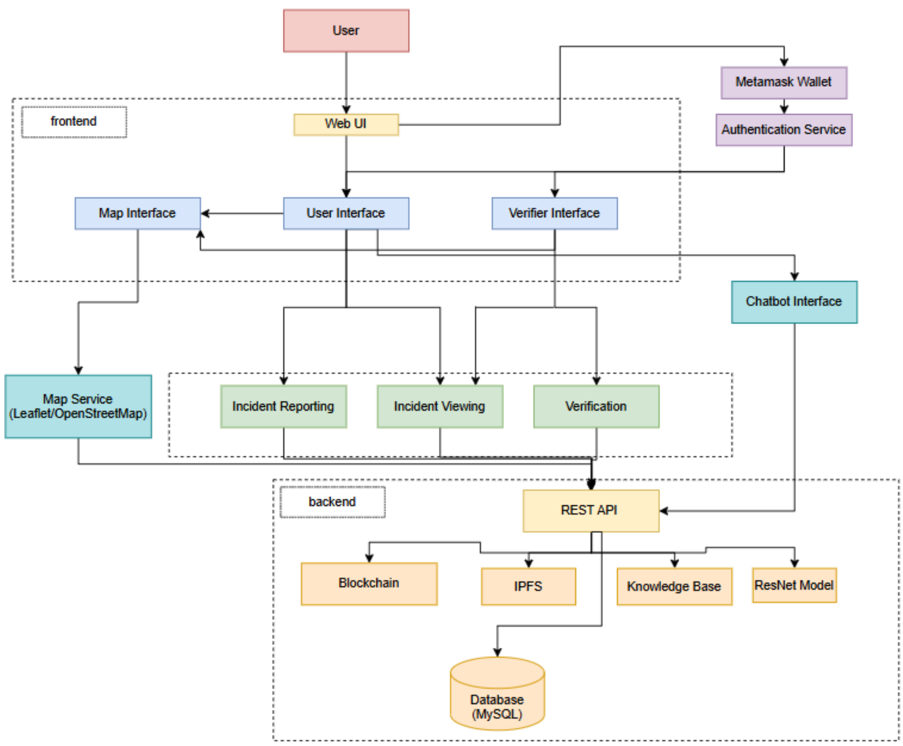
ARCHITECTURE DIAGRAM



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MODULE-WISE DESCRIPTION

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INCIDENT REPORTING MODULE

- Allows users to submit incident reports through a web interface where they can provide descriptions, geolocation coordinates, timestamps, and image evidence.
- The frontend communicates with the backend using Node.js and Web3 to process incident data and interact with the blockchain network.
- Uploaded images are attached to the report and later stored using decentralized storage for verification and transparency.

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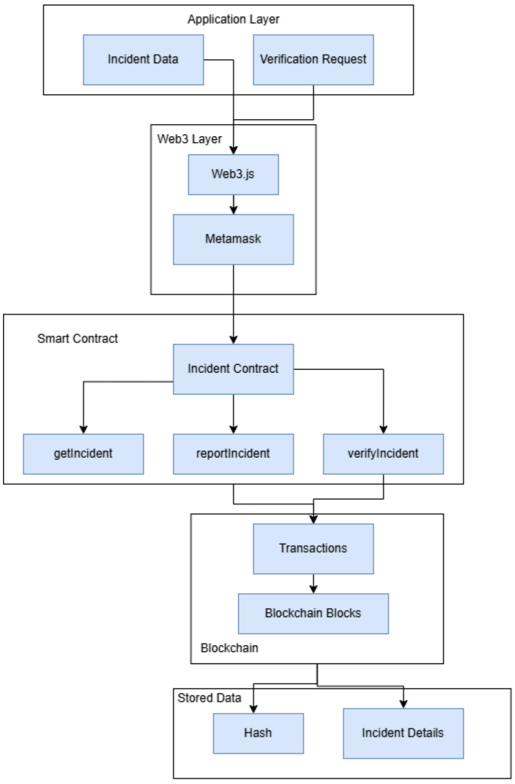

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BLOCKCHAIN MODULE

- The system uses smart contracts to manage incident registration, verification logic, and access control.
- Smart contracts are developed and deployed using the Truffle framework and tested in a local blockchain environment using Ganache.
- User authentication and transaction signing are performed through MetaMask wallets integrated with Web3.js to ensure secure and tamper-proof records.

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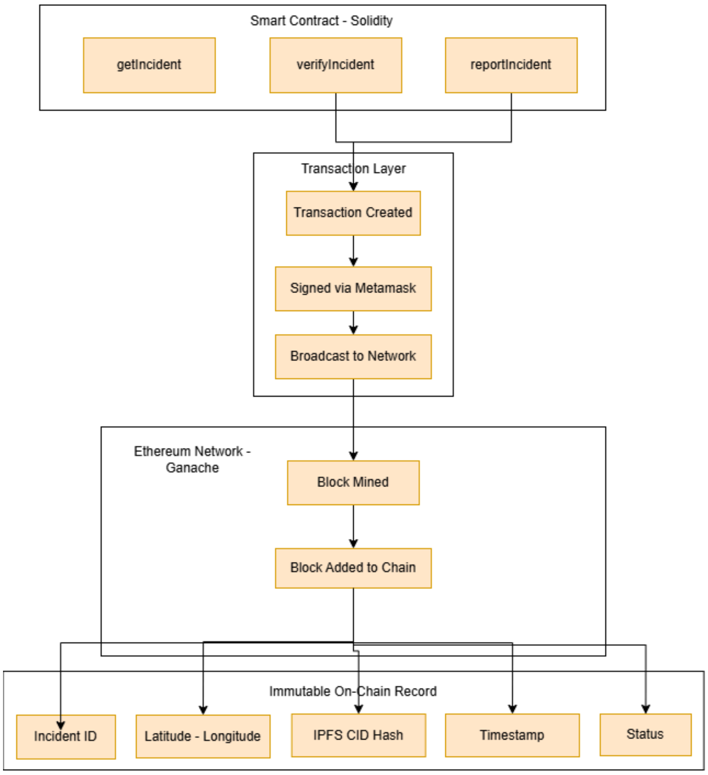

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MAP-BASED VISUALIZATION MODULE

- Reported incidents are displayed on an interactive map implemented using Leaflet.js with OpenStreetMap for geographic visualization.
- Incidents appear as markers placed using latitude and longitude coordinates retrieved from the system database and blockchain records.
- Color-coded markers help users easily distinguish between verified and unverified incidents and view detailed information when clicked.

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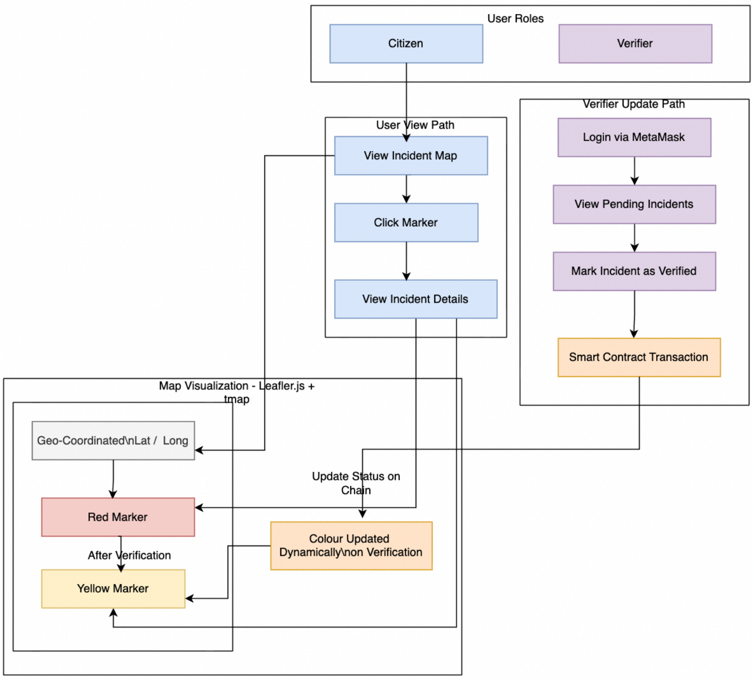

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OFF-CHAIN STORAGE MODULE

- Incident descriptions and related metadata are stored in a MySQL relational database to allow efficient querying and retrieval of incident information.

- Images uploaded during reporting are stored in IPFS (InterPlanetary File System) using the Pinata API, which generates a unique Content Identifier (CID).

- These database records are linked to blockchain transactions using unique identifiers and IPFS hashes to maintain data integrity.

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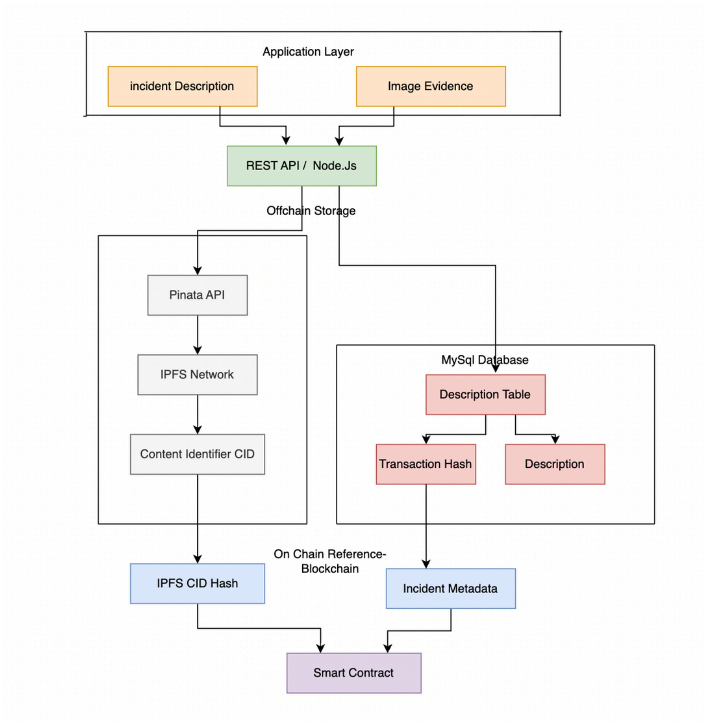

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INCIDENT VERIFICATION MODULE

- Authorized verifiers review submitted incidents through a verification interface connected to the blockchain using Web3.js.
- When a verifier confirms an incident, a new blockchain transaction updates the verification status through the Solidity smart contract.
- The updated status is immediately reflected in the incident list and the Leaflet-based map visualization, improving reliability of reported information.

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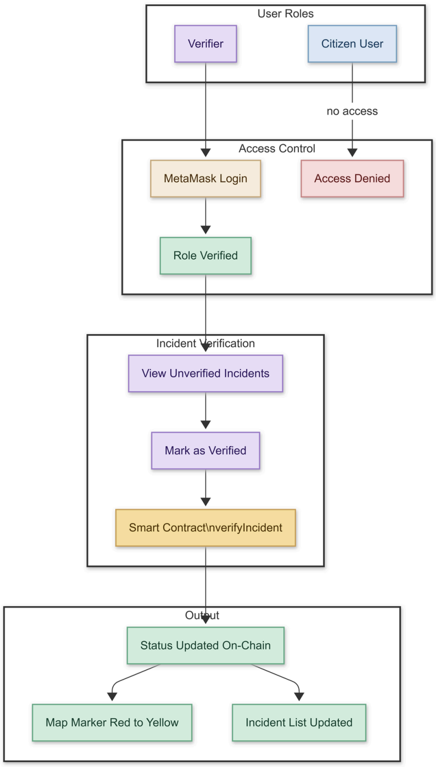
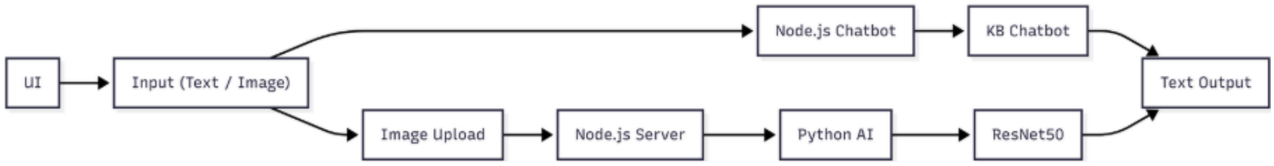


CHATBOT ASSISTANCE MODULE

- An AI-powered chatbot integrated into the web interface helps users understand the system and guides them through the incident reporting process.
- The chatbot can analyze uploaded images using an AI image analysis model (such as ResNet) connected to a knowledge base to identify the incident type and severity.
- It also answers user queries and suggests precautionary measures, improving user interaction and awareness.



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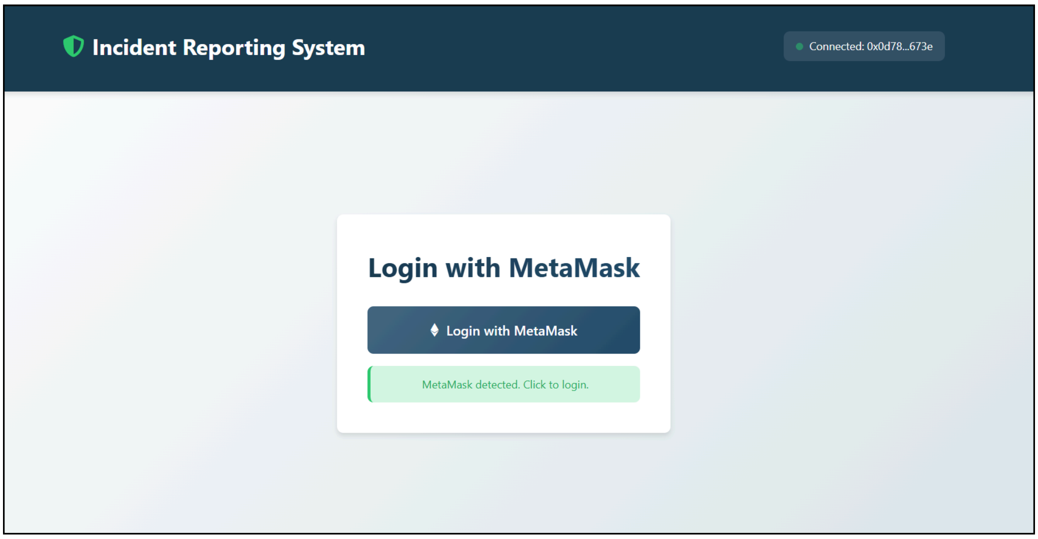
RESULT AND OUTPUTS



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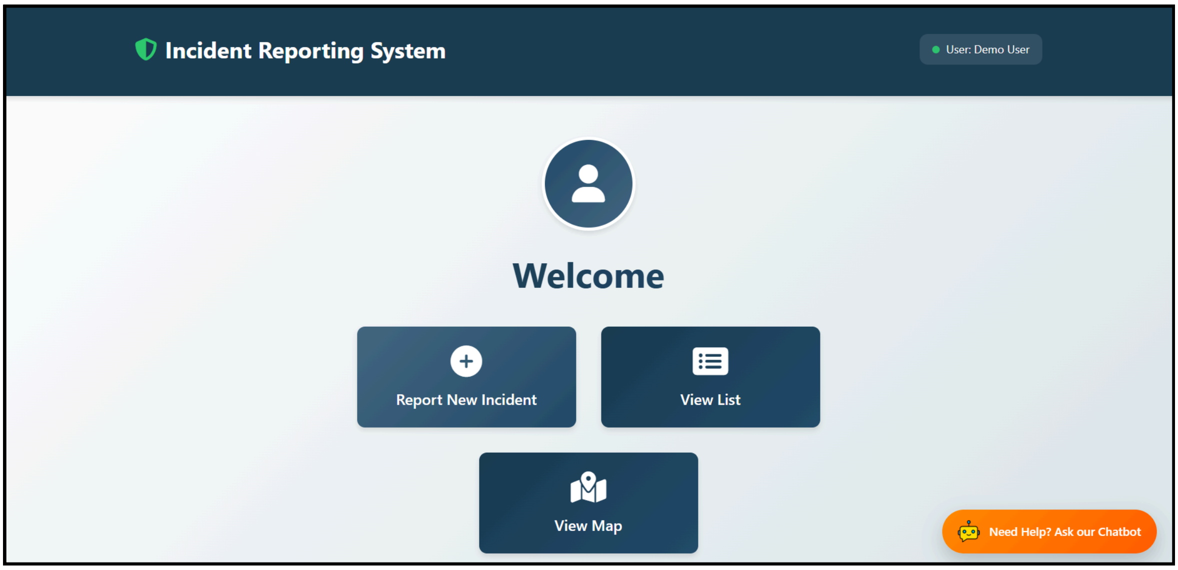

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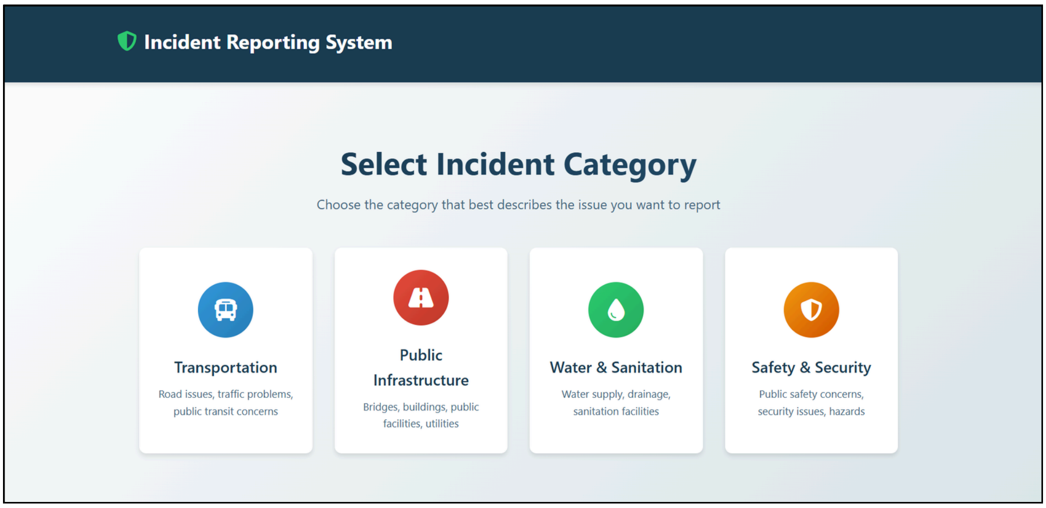

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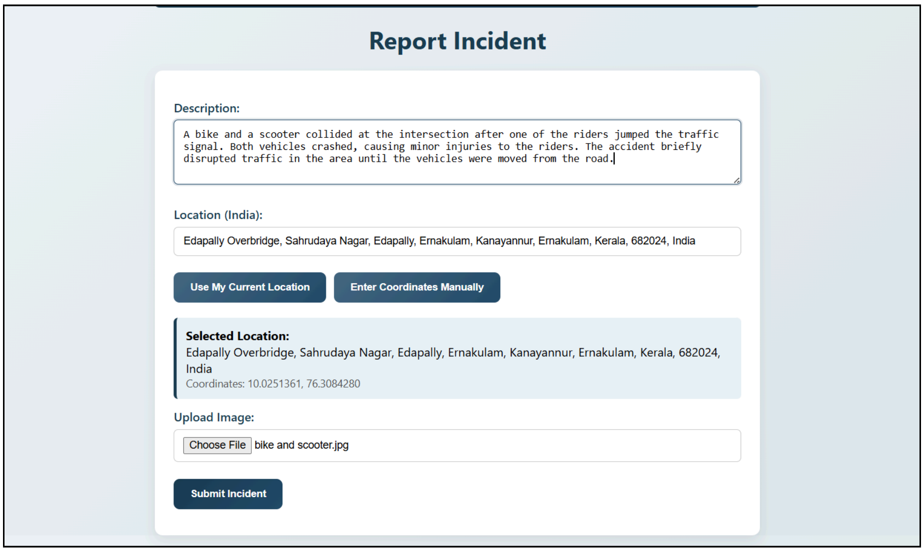

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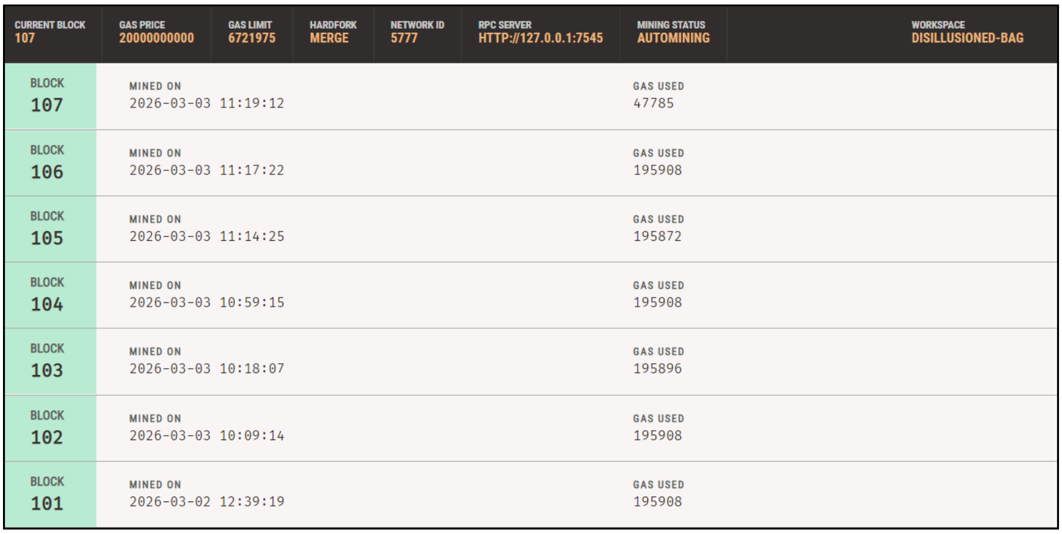

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PERFORMANCE EVALUATION OF INCIDENT SUBMISSION

Incident Type	IPFS Time (s)	Blockchain Time (s)	Total Submission Time (s)
Transportation	2.508	4.142	8.401
Infrastructure	3.186	4.665	9.453
Water & Sanitation	1.733	2.828	6.248
Safety & Security	3.782	2.961	8.584

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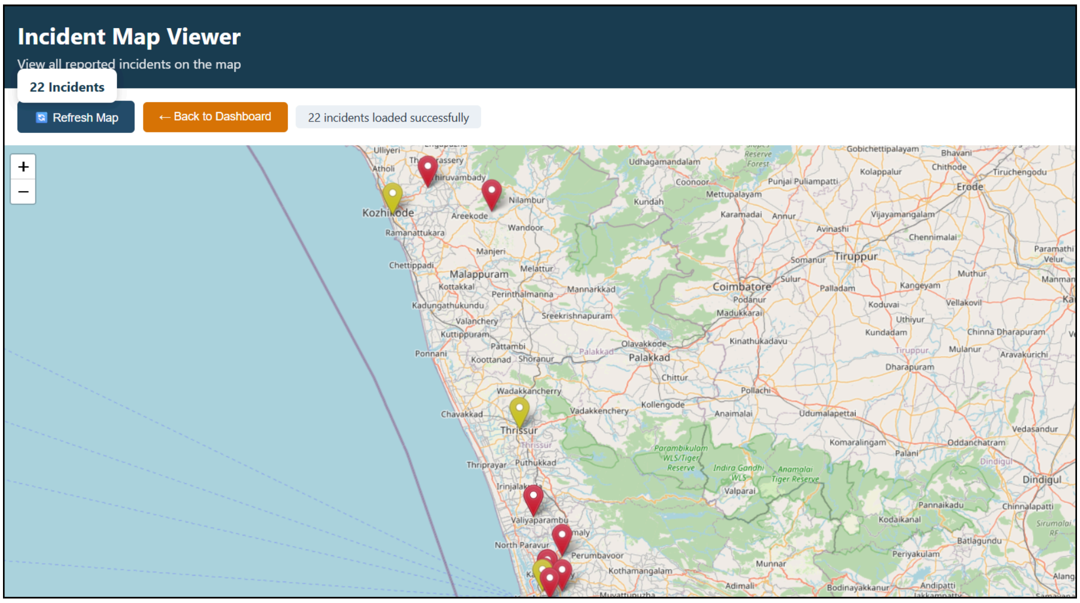

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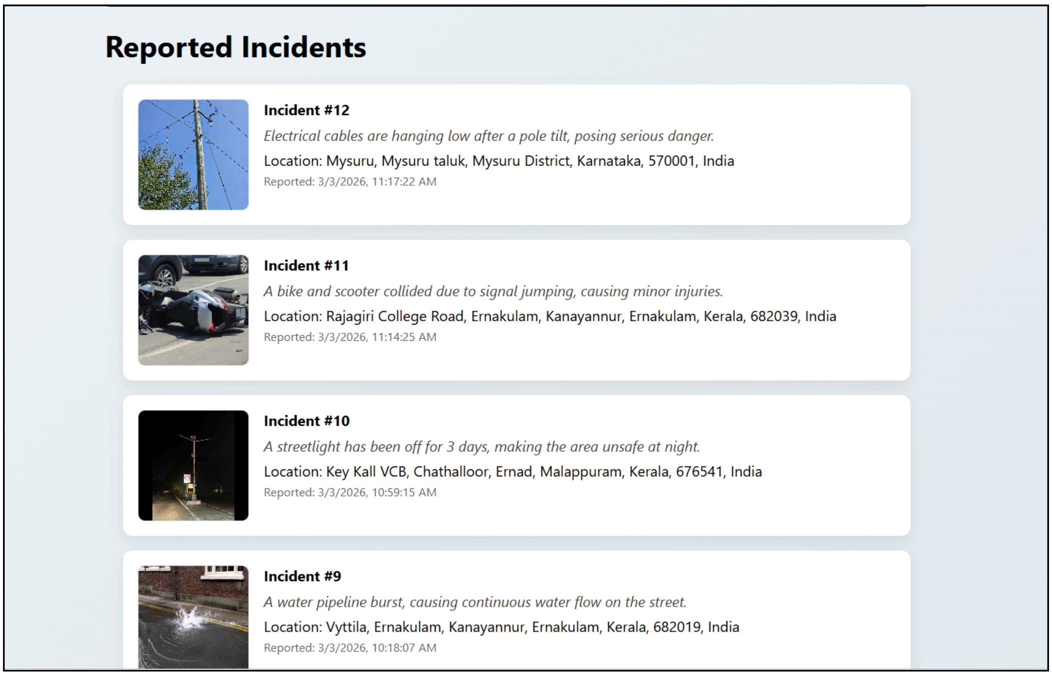

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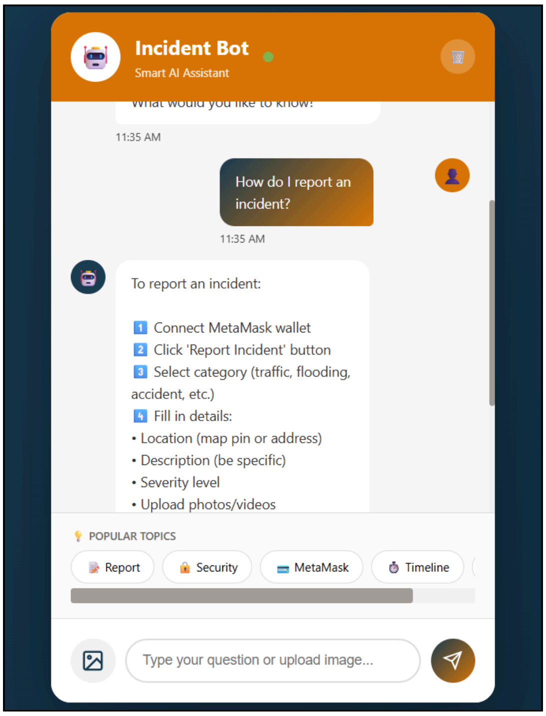

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AI Description

Road Damage/Pothole detected with 100.0% confidence using ONNX AI model.

Analysis Breakdown

Image Quality	100%
Object Detection	100%
Context Relevance	95%
Manipulation Detection	95%

Recommended Steps

1. ⚠️ If safe, mark hazard area with cones, branches, or warning objects
2. Measure or estimate: Length, width, depth of pothole/damage
3. Clear photos with reference object (coin, phone, ruler) for scale

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Incident #16

open drain slab

Location: Vazha Paramp, Varattiakkal-Manipuram Road, Abalaparambil Colony, Padanilam, Kozhikode, Kerala, 673601, India

Reported: 3/8/2026, 7:27:39 PM

Mark as Verified

Incident #15

contaminated water supply

Location: Perambur, Perambur High Road, CMWSSB Division 71, Ward 71, Zone 6 Thiru. Vi. Ka. Nagar, Chennai, Tamil Nadu, 600011, India

Reported: 3/8/2026, 7:19:09 PM

Mark as Verified

Incident #14

No description available

Location: Location not available

Reported: 3/8/2026, 7:04:37 PM

Verified

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ASSUMPTIONS

- Users will have access to a stable internet connection and a modern web browser.
- Citizens will provide accurate and truthful incident details.
- Users will have basic familiarity with web platforms and location-sharing features.

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WORK BREAKDOWN AND RESPONSIBILTY

Susen

- Design and implement smart contracts in Solidity
- Train the chatbot to answer user queries and correctly identify incident images
- Work on image-based incident description and precaution suggestions
- Writing and documentation of project reports

Siona

- Handle incident creation, storage of on-chain data and verification logic
- Map-based visualization using Leaflet and OpenStreetMap
- Manage off-chain storage using a database for incident descriptions
- Handle IPFS integration for image uploads and retrieval
- Final integration of all modules and testing of system

Treesa

- Map-based visualization using Leaflet and OpenStreetMap
- Implement incident verification with dynamic map marker updates
- Writing and documentation of project reports

Niya

- Design and implement the web UI using HTML and CSS
- Design and preparation of project presentations
- Writing and documentation of project reports

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SOFTWARE REQUIREMENTS

- Solidity (v0.8.19)
- Truffle (v5.11.5)
- Ganache (v7.9.1)
- Web3.js (v1.10.0)
- MetaMask
- IPFS
- Leaflet (Map API)
- HTML / CSS / JavaScript

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HARDWARE REQUIREMENTS

- Minimum RAM: 4 GB
- Recommended RAM: 8 GB
- Minimum Processor: Dual-core CPU
- Recommended Processor: Quad-core CPU
- Internet: Stable / high-speed connection
- Browser: Latest Google Chrome or Mozilla Firefox
- Wallet: MetaMask for blockchain authentication
- Storage: Basic local storage for caching and app data

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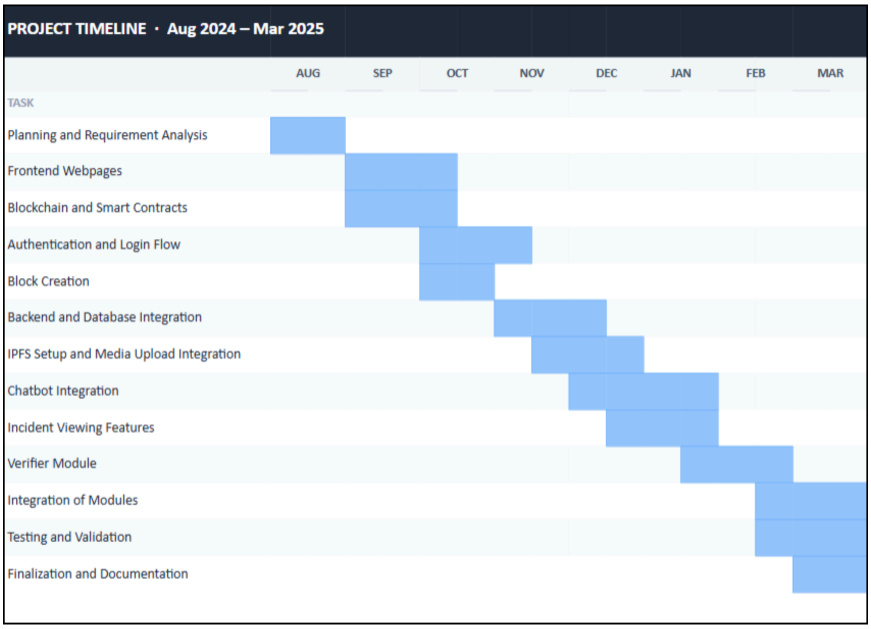
RISK AND CHALLENGES

- **Blockchain Network Issues:** Transaction delays, gas fees, or network congestion may affect incident reporting and verification speed.
- **User Adoption:** Users unfamiliar with MetaMask and cryptocurrency wallets may face difficulties in accessing the system.
- **Fake Incident Reports:** Malicious users may submit false or misleading incidents, requiring robust verification processes to maintain data credibility.

- **Scalability and Cost:** Increasing number of incidents leads to higher blockchain storage costs and transaction fees over time.
- **Verifier Accountability:** Risk of malicious or biased verifiers incorrectly approving/rejecting incidents without proper oversight mechanisms.

GANTT CHART



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CONCLUSION

- Decentralization removes reliance on a single authority and improves public trust.
- Real-time incident reporting and map visualization enhance urban safety and awareness.
- Smart contracts ensure secure, automated, and verifiable incident handling.
- The project demonstrates the practical use of blockchain for smart city governance.

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THANK YOU

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Appendix B:

Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: Engineering Knowledge Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions to complex engineering problems.

PO2: Problem Analysis Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions Design creative solutions for complex engineering problems and design/develop systems, components or processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modelling while recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

PO8: Individual and Collaborative Team Work Function effectively as an individual and as a member or leader in diverse and multidisciplinary teams.

PO9: Communication Communicate effectively and inclusively within the engineering community and society at large.

PO10: Project Management and Finance Apply knowledge and understanding of engineering management principles and economic decision-making to manage projects in multidisciplinary environments.

PO11: Life-Long Learning Recognize the need for and have the preparation and ability for independent and life-long learning, adaptability to new and emerging technologies and critical thinking.

Programme Specific Outcomes (PSO)

PSO1: Computer Science Specific Skills

Ability to identify, analyze, and design solutions for complex engineering problems by understanding the core principles of computer science.

PSO2: Programming and Software Development Skills

Ability to design algorithms and apply software development practices to deliver quality software products.

PSO3: Professional Skills

Ability to apply computer science fundamentals in research and develop innovative solutions to meet societal needs.

Course Outcomes (CO)

After completion of the course the student will be able to

SL.NO	DESCRIPTION
CO1	Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).
CO2	Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).
CO3	Function effectively as an individual and as a leader in diverse teams (Cognitive knowledge level: Apply).
CO4	Plan and execute tasks utilizing available resources within timelines following ethical and professional norms (Cognitive knowledge level: Apply).
CO5	Identify technology or research gaps and propose innovative solutions (Cognitive knowledge level: Analyze).
CO6	Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	**Level**	**Justification**
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.

Mapping	Level	Justification
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.

Mapping	Level	Justification
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.

Mapping	Level	Justification
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.

Mapping	Level	Justification
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.